Supplementary Material

Zero-Sample Fault Diagnosis for EHSS: A Hierarchical Multi-Source Hybrid PI-RNN Based Method

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S1 Complete Model Supplement and Parameters

This supplementary material provides the detailed physical model of the electrohydraulic servo system (EHSS) presented in the main manuscript. It specifies the function for friction, leakage, valve flow, and electromagnetic actuation, together with the governing state equations and numerical implementation. Key physical parameters are summarized in Table S1.

Friction, flow force, and leakage. The spool and actuator friction terms are

$$\begin{aligned} F_{\mathrm{fric},s} &= -B_s\dot{x}_s - \left(F_{Cs} + F_{Ss}e^{-(\dot{x}_s/v_{Ss})^2}\right)\tanh\left(\frac{\dot{x}_s}{v_{\epsilon,s}}\right), \\ F_{\mathrm{fric},c} &= B_c\dot{x}_c + \left(F_{Cc} + F_{Sc}e^{-(\dot{x}_c/v_{Sc})^2}\right)\tanh\left(\frac{\dot{x}_c}{v_{\epsilon,c}}\right) \\ &\quad + B_{c,\mathrm{asym}}\dot{x}_c\tanh(\dot{x}_c/v_{\epsilon,c}) \\ &\quad + F_{Cc,\mathrm{asym}}\tanh^2(\dot{x}_c/v_{\epsilon,c}). \end{aligned} \tag{S1}$$

$$B_s = 2.46, \quad B_c = 8.96\times10^{3}, \quad B_{c,\mathrm{asym}} = 6.68\times10^{3}\ \mathrm{N\,s/m},$$

$$F_{Cs} = 1.17\times10^{-3}, \quad F_{Ss} = 6.09\times10^{-3}, \quad F_{Cc} = 746, \quad F_{Sc} = 97.5, \quad F_{Cc,\mathrm{asym}} = 702\ \mathrm{N},$$

$$v_{Ss} = 0.0105, \quad v_{\epsilon,s} = 0.00243, \quad v_{Sc} = 0.00729, \quad v_{\epsilon,c} = 0.00851\ \mathrm{m/s}.$$

The pressure-dependent force acting on the spool is

$$\begin{aligned} F_{\mathrm{flow},s} = -\Bigg[&c_{F,1}\frac{P_A - P_B}{P_{\mathrm{ref}}} + c_{F,2}\frac{P_A + P_B}{P_{\mathrm{ref}}} \\ &+ c_{F,3}\frac{P_A^2 - P_B^2}{P_{\mathrm{ref}}^2}\Bigg]. \end{aligned} \tag{S2}$$

$$(c_{F,1}, c_{F,2}, c_{F,3}) = (0.0378, 0.0568, -0.0253)\ \mathrm{N}, \qquad P_{\mathrm{ref}} = 6.89\times10^{6}\ \mathrm{Pa}.$$

The cross-chamber leakage is

$$Q_{L,A} = C_\ell(P_A - P_B), \qquad Q_{L,B} = -C_\ell(P_A - P_B). \tag{S3}$$

$$C_\ell = 8.49\times10^{-13}\ \mathrm{m^3/(s\,Pa)}.$$

Metering areas and discharge coefficients. The metering areas are defined using $\mathcal{A}(w, z) = \min\{A_{\max}, w \min(\ell_u, [z]_+)\}$, where $[z]_+ = \max(z, 0)$:

$$\begin{aligned} A_A &= \mathcal{A}(w_{SA}, x_s/s_x - d_A), \\ A_{A2} &= \mathcal{A}(w_{TA}, -x_s/s_x - d_A), \\ A_B &= \mathcal{A}(w_{SB}, -x_s/s_x - d_B), \\ A_{B2} &= \mathcal{A}(w_{TB}, x_s/s_x - d_B). \end{aligned} \tag{S4}$$

$$(w_{SA}, w_{TA}, w_{SB}, w_{TB}) = (13.9, 11.6, 6.92, 11.3) \times 10^{-3}\,\mathrm{m}, \quad (d_A, d_B) = (7.38, 6.45) \times 10^{-6}\,\mathrm{m},$$
$$A_{\max} = 1.13 \times 10^{-5}\,\mathrm{m}^2, \qquad s_x = 1, \qquad \ell_u = 4.92 \times 10^{-4}\,\mathrm{m}.$$

Both edges of chamber $j \in \{A, B\}$ use the discharge coefficient

$$\begin{aligned} C_{d,j} = \mathrm{clip}_{[C_{d,\min}, C_{d0}]}\Big\{ & \\ & C_{d0}\Big[1 + k_o \min\Big(\frac{|x_s|}{s_x \ell_u}, 1\Big) \\ & \quad + k_{Re} \frac{Re_j}{Re_0 + Re_j}\Big]\Big\}, \\ Re_j = \frac{4\rho|Q_j|}{\pi \mu D_h}. & \end{aligned} \tag{S5}$$

$$(C_{d,\min}, C_{d0}) = (0.300, 0.650), \quad (k_o, k_{Re}) = (-0.397, 0.169), \quad Re_0 = 2.00 \times 10^3, \quad D_h = 0.0120\,\mathrm{m}.$$

Pressure-dependent flow is modeled by

$$\begin{aligned} \mathcal{R}_{n_q}(\Delta P) = & \sqrt{\frac{2P_{\mathrm{ref}}}{\rho}} \frac{\Delta P}{P_{\mathrm{ref}}} \\ & \times \left[\left(\frac{\Delta P}{P_{\mathrm{ref}}}\right)^2 + \left(\frac{\epsilon_p}{P_{\mathrm{ref}}}\right)^2\right]^{n_q/4 - 1/2}. \end{aligned} \tag{S6}$$

$$n_q = 0.703, \qquad \epsilon_p = 1.00 \times 10^4\,\mathrm{Pa}.$$

Physical-state equations. With $v_s = \dot{x}_s$ and $v_c = \dot{x}_c$, the nine-state model is

$$\begin{aligned} \dot{i} &= \frac{u - Ri - \lambda_x v_s}{\lambda_i}, \qquad \dot{x}_s = v_s, \\ \dot{v}_s &= \frac{1}{m_s}\big(F_{em} + F_{\mathrm{flow},s} - F_{s0} - k_s x_s \\ &\qquad + F_{\mathrm{fric},s} + S_s\big), \\ \dot{P}_A &= \frac{\beta_e(Q_A + A_r v_c - Q_{L,A})}{V_A(x_c)}, \\ \dot{P}_B &= \frac{\beta_e(Q_B - A_p v_c - Q_{L,B})}{V_B(x_c)}, \\ \dot{x}_c &= v_c, \\ \dot{v}_c &= \frac{1}{m_c}\big(A_p P_B - A_r P_A - F_{\mathrm{fric},c} \\ &\qquad + S_c - F_{ext}\big). \end{aligned} \tag{S7}$$

The flow dynamics are

$$\begin{aligned}\dot{Q}_A &= \frac{1}{L_{f,A}}\Big\{C_{d,A}\big[A_A\mathcal{R}_{n_q}(P_S - P_A) \\ &\qquad - A_{A2}\mathcal{R}_{n_q}(P_A - P_T)\big] - R_{f,A}Q_A\Big\}, \\ \dot{Q}_B &= \frac{1}{L_{f,B}}\Big\{C_{d,B}\big[A_B\mathcal{R}_{n_q}(P_S - P_B) \\ &\qquad - A_{B2}\mathcal{R}_{n_q}(P_B - P_T)\big] - R_{f,B}Q_B\Big\}. \\ (R_{f,A}, R_{f,B}) &= (1.08, 0.568).\end{aligned} \tag{S8}$$

The state-dependent chamber volumes are

$$\begin{aligned}V_A(x_c) &= \max\left\{q_V V_{\text{rod,dead}}, V_{A0} - A_r x_c\right\}, \\ V_B(x_c) &= \max\left\{q_V V_{B0}, V_{B0} + A_p x_c\right\}.\end{aligned} \tag{S9}$$

$$q_V = 0.250.$$

The reference position is fully retracted. To enforce nonnegative pressure, negative rates at zero pressure are suppressed and negative values after integration are projected to zero.

The signed mechanical stop forces are

$$\begin{aligned}S_s &= -k_{\text{stop},s}\,\text{sign}(x_s)\,[|x_s| - x_{s,\max}]_+ \\ &\quad - b_{\text{stop},s}v_s\mathbf{1}_{|x_s|>x_{s,\max}}, \\ S_c &= -k_{\text{stop},c}\left[x_c - \text{clip}_{[0,\ell_c]}(x_c)\right] \\ &\quad - b_{\text{stop},c}\Big[\min(v_c, 0)\mathbf{1}_{x_c<0} \\ &\qquad + \max(v_c, 0)\mathbf{1}_{x_c>\ell_c}\Big].\end{aligned} \tag{S10}$$

The contact settings are $x_{s,\max} = 2.00 \times 10^{-3}$ m, $(k_{\text{stop},s}, k_{\text{stop},c}) = (2.00 \times 10^6, 1.50 \times 10^8)$ N/m, and $(b_{\text{stop},s}, b_{\text{stop},c}) = (1.00 \times 10^2, 9.95 \times 10^4)$ N s/m. The actuator restoring force is S_c. The unloaded simulation uses $F_{ext} = 0$.

Electrical and electromagnetic fits.

$$
\begin{aligned}
\Phi_0 &= a(i_e) + g_e b(i_e) + g_e^2 c(i_e),\\
\lambda &= N_c \Phi_0,\\
i_e &= \mathrm{clip}_{[0.800,\,2.85]\,\mathrm{A}}(i),\\
g_e &= \mathrm{clip}_{[0,\,4.00\times10^{-4}]\,\mathrm{m}}\left(2.12\times10^{-4}\,\mathrm{m} - x_s\right),\\
a(i_e) &= \sum_{n=0}^{9} a_{\Phi,n} i_e^n,\\
b(i_e) &= \sum_{n=0}^{8} b_{\Phi,n} i_e^n,\\
c(i_e) &= \sum_{n=0}^{7} c_{\Phi,n} i_e^n,\\
F_{em} &= \sum_{p,q} e_{pq} i_e^p \Phi_0^q,\\
\lambda_i &= N_c\left[a'(i_e) + g_e b'(i_e) + g_e^2 c'(i_e)\right],\\
\lambda_x &= -N_c\left[b(i_e) + 2 g_e c(i_e)\right]\\
&\quad \times \mathbf{1}_{0<2.12\times10^{-4}\,\mathrm{m}-x_s<4.00\times10^{-4}\,\mathrm{m}}.
\end{aligned}
\tag{S11}
$$

$$
a_{\Phi,n},\ b_{\Phi,n},\ c_{\Phi,n},\ e_{pq}
$$

are provided in the accompanying parameter file; all unspecified terms are zero.

Table S1: Nominal physical parameters of the EHSS model.

Parameter	Symbol	Value (SI unit)
Coil resistance	R	$3\,\Omega$
Coil turns	N_c	600
Spool mass	m_s	$0.15\,\mathrm{kg}$
Actuator mass	m_c	$12\,\mathrm{kg}$
Spring preload	F_{s0}	$5\,\mathrm{N}$
Spool spring stiffness	k_s	$2000\,\mathrm{N/m}$
Rod-end piston area	A_r	$2.16\times10^{-3}\,\mathrm{m}^2$
Cap-end piston area	A_p	$3.12\times10^{-3}\,\mathrm{m}^2$
Actuator stroke	ℓ_c	$0.8\,\mathrm{m}$
Supply pressure	P_S	$6.89\times10^{6}\,\mathrm{Pa}$
Tank gauge pressure	P_T	$0\,\mathrm{Pa}$
Oil density	ρ	$850\,\mathrm{kg/m}^3$
Oil dynamic viscosity	μ	$0.0391\,\mathrm{Pa\,s}$
Effective bulk modulus	β_e	$1.24\times10^{9}\,\mathrm{Pa}$
Initial rod-end volume	V_{A0}	$1.75\times10^{-3}\,\mathrm{m}^3$
Initial cap-end volume	V_{B0}	$1.62\times10^{-3}\,\mathrm{m}^3$
Rod-side dead volume	$V_{\mathrm{rod,dead}}$	$2.93\times10^{-5}\,\mathrm{m}^3$

Physical parameters. Table S1 lists nominal system parameters.

S2 Convolution-kernel design and disturbance verification

The EHSS is characterized by a cascaded physical structure, following $u \to i \to x_s \to \{P_A, P_B\} \to x_c \to F_{\text{ext}}$. Based on these physical dependencies, the convolutional kernels $\boldsymbol{K}_S$ and $\boldsymbol{K}_C$ are designed through a physics-constrained optimization procedure, as detailed below.

The main structure is shown in the Figure S1. The three physical stages are the proportional-solenoid

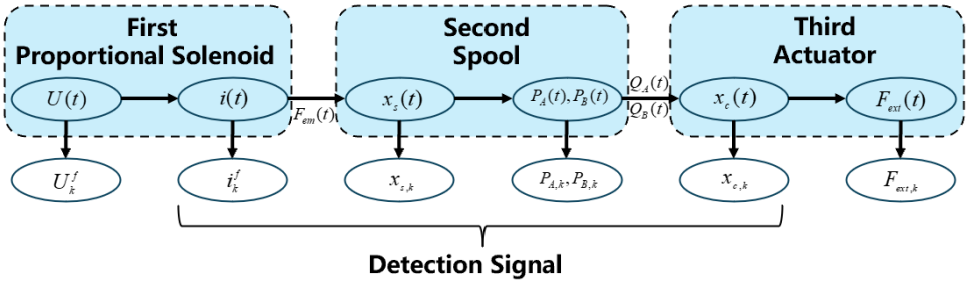


Fig. S1: Hierarchical Structure and Signal Definitions.

stage, the spool stage, and the hydraulic-actuator stage. They describe subsystem decomposition and are distinct from the signal nodes used in the hierarchical residual construction. The four internal monitored hierarchy levels are

$$\mathcal{H}_1 = \{i\}, \qquad \mathcal{H}_2 = \{x_s\}, \qquad \mathcal{H}_3 = \{\boldsymbol{P}\}, \qquad \mathcal{H}_4 = \{x_c\}, \quad \boldsymbol{P} = [P_A, P_B]^{\mathrm{T}}. \tag{S12}$$

The input u and the load variable F_{ext} are non-diagnostic boundary/context nodes. Hence, the grouped topological sequence used for hierarchical inference is

$$[u,\ i,\ x_s,\ \boldsymbol{P},\ x_c,\ F_{\text{ext}}], \qquad N_h = 6, \quad \mathcal{V} = \{1, \ldots, N_h\}, \quad \mathcal{V}_M = \{2, \ldots, 5\}. \tag{S13}$$

Here, $N_h = 6$ counts grouped topological nodes, whereas N_c is reserved for the number of solenoid coil turns. The set $\mathcal{V}$ denotes hierarchy nodes and is distinct from the diagnostic indicator $\mathcal{I}$. The four internal nodes contain five monitored scalar channels, $[i, x_s, P_A, P_B, x_c]^{\mathrm{T}}$, because the pressure node contains two chamber-pressure components. In particular, P_A and P_B are parallel components of the same hydraulic level; neither is upstream or downstream of the other.

S2.1 Kernel-design procedure

At each hierarchy level, three types of information are available: upstream inference, PI-RNN prediction, and downstream inference. The pressure pair (P_A, P_B) occupies one physical level. As in the manuscript, the normalized deviation matrix is formed separately for pressure branches $b \in \{A, B\}$; the same 3×3 kernels are applied to both branches. After saturation and threshold processing, it is written as

$$\bar{\boldsymbol{D}}_{\nu,k}^{(b)} = \frac{a_{\text{sat}} |\boldsymbol{D}_{\nu,k}^{(b)}|}{a_{\text{sat}} + |\boldsymbol{D}_{\nu,k}^{(b)}|}, \tag{S14}$$

$$\widetilde{\boldsymbol{D}}_{\nu,k}^{(b)}(i,j) = \begin{cases} \bar{\boldsymbol{D}}_{\nu,k}^{(b)}(i,j), & \bar{\boldsymbol{D}}_{\nu,k}^{(b)}(i,j) > \frac{a_{\text{sat}}}{a_{\text{sat}}+1}, \\ 0, & \bar{\boldsymbol{D}}_{\nu,k}^{(b)}(i,j) \le \frac{a_{\text{sat}}}{a_{\text{sat}}+1}. \end{cases} \tag{S15}$$

Here $a_{\mathrm{sat}} = 4$, so the activation cutoff is 0.8. For each branch, valid convolution with unit stride and no padding extracts four local residual windows:

$$\mathbf{U}^{(b)}_{\nu,k,l} = \widetilde{\boldsymbol{D}}^{(b)}_{\nu,k}(l : l+2, :) \in \mathbb{R}^{3\times 3}, \qquad l = 1, \ldots, 4. \tag{S16}$$

The branch-specific diagnostic responses are defined by

$$z^{(b)}_{S,k,l} = \langle \mathbf{U}^{(b)}_{S,k,l}, \boldsymbol{K}_S \rangle, \qquad z^{(b)}_{C,k,l} = \langle \mathbf{U}^{(b)}_{C,k,l}, \boldsymbol{K}_C \rangle, \tag{S17}$$

where $\boldsymbol{K}_S$ is the sensor-fault kernel, $\boldsymbol{K}_C$ is the component-propagation kernel, and $\langle \cdot, \cdot \rangle$ denotes the Frobenius inner product. For $\nu \in \{S, C\}$, define $\bar{z}_{\nu,k,l} = (z^{(A)}_{\nu,k,l} + z^{(B)}_{\nu,k,l})/2$. The two pressure branches are fused exactly as in the manuscript:

$$\mathbf{z}_{\nu,k} = [\bar{z}_{\nu,k,1}, \bar{z}_{\nu,k,2}, z^{(A)}_{\nu,k,3}, z^{(B)}_{\nu,k,3}, \bar{z}_{\nu,k,4}]^{\mathrm{T}}. \tag{S18}$$

Its five entries correspond to $[i, x_s, P_A, P_B, x_c]^{\mathrm{T}}$. The first, second, and fifth responses are averaged across branches, while the two pressure responses remain separate. In the design inequalities below, $z^{S}_{k,l}$ and $z^{C}_{k,l}$ denote the entries of these final fused indicators, with $l = 1, \ldots, 5$.

The kernel optimization starts from the physical residual patterns of the EHSS. For a sensor fault, the physical state is unchanged, while the local measurement becomes inconsistent with upstream inference, PI-RNN prediction, and downstream inference. Thus, the expected sensor-fault residual topology is

$$\mathbf{M}_S = \begin{bmatrix} 0 & 0 & + \\ + & + & + \\ + & 0 & 0 \end{bmatrix}. \tag{S19}$$

For a component fault, the physical state changes and the residual propagates downstream along the causal chain. Thus, the expected component-fault residual topology is

$$\mathbf{M}_C = \begin{bmatrix} 0 & 0 & + \\ + & + & 0 \\ 0 & + & 0 \end{bmatrix}. \tag{S20}$$

Based on these two physical templates, paired A/B-branch residual-pattern datasets are generated for kernel selection. In the following dataset notation, $\mathbf{W}$ represents the branch-window pair $\{\mathbf{W}^{(A)}, \mathbf{W}^{(B)}\}$ and its associated fused five-position response:

$$\mathcal{D}_N = \{\mathbf{W}^{N}_{k,l}\}, \tag{S21}$$

$$\mathcal{D}_S = \{(\mathbf{W}^{S}_{k,l}, l^*)\}, \tag{S22}$$

$$\mathcal{D}_C = \{(\mathbf{W}^{C}_{k,l}, l^*)\}, \tag{S23}$$

where $\mathcal{D}_N$, $\mathcal{D}_S$, and $\mathcal{D}_C$ denote normal, sensor-fault, and component-fault residual windows, respectively, and l^* is the true fault layer.

The optimization objective is to find two kernels that satisfy the following diagnostic inequalities.

For normal samples:

$$z^S_{k,l} < \tau_S, \qquad z^C_{k,l} < \tau_C, \qquad \forall l. \tag{S24}$$

For a sensor fault at layer l^*:

$$z^S_{k,l^*} > \tau_S, \tag{S25}$$

$$z^S_{k,l} < \tau_S, \qquad l \neq l^*, \tag{S26}$$

$$z^C_{k,l} < \tau_C, \qquad \forall l. \tag{S27}$$

For a component fault starting at layer l^*:

$$z^C_{k,l} > \tau_C, \qquad l \geq l^*, \tag{S28}$$

$$z^C_{k,l} < \tau_C, \qquad l < l^*, \tag{S29}$$

$$z^S_{k,l} < \tau_S, \qquad \forall l. \tag{S30}$$

In this work, the thresholds are set as

$$\tau_S = 4, \qquad \tau_C = 5. \tag{S31}$$

The above inequalities are converted into a hinge-loss optimization problem:

$$\begin{aligned} \min_{\boldsymbol{K}_S, \boldsymbol{K}_C} \quad & \mathcal{L}_N + \mathcal{L}_S + \mathcal{L}_C + \lambda_1 \left(\|\boldsymbol{K}_S\|_1 + \|\boldsymbol{K}_C\|_1 \right) \\ & + \lambda_2 \left(\|\boldsymbol{K}_S\|_F^2 + \|\boldsymbol{K}_C\|_F^2 \right), \end{aligned} \tag{S32}$$

where the normal-condition loss is

$$\mathcal{L}_N = \sum_{\mathbf{W} \in \mathcal{D}_N} \sum_{l} \left[z^S_{k,l} - \tau_S + \eta_S \right]_+^2 + \left[z^C_{k,l} - \tau_C + \eta_C \right]_+^2. \tag{S33}$$

The sensor-fault loss is

$$\begin{aligned} \mathcal{L}_S = & \sum_{(\mathbf{W}, l^*) \in \mathcal{D}_S} \left[\tau_S + \eta_S - z^S_{k,l^*} \right]_+^2 \\ & + \sum_{(\mathbf{W}, l^*) \in \mathcal{D}_S} \sum_{l \neq l^*} \left[z^S_{k,l} - \tau_S + \eta_S \right]_+^2 \\ & + \sum_{(\mathbf{W}, l^*) \in \mathcal{D}_S} \sum_{l} \left[z^C_{k,l} - \tau_C + \eta_C \right]_+^2. \end{aligned} \tag{S34}$$

The component-fault loss is

$$\begin{aligned} \mathcal{L}_C = & \sum_{(\mathbf{W}, l^*) \in \mathcal{D}_C} \sum_{l \geq l^*} \left[\tau_C + \eta_C - z^C_{k,l} \right]_+^2 \\ & + \sum_{(\mathbf{W}, l^*) \in \mathcal{D}_C} \sum_{l < l^*} \left[z^C_{k,l} - \tau_C + \eta_C \right]_+^2 \\ & + \sum_{(\mathbf{W}, l^*) \in \mathcal{D}_C} \sum_{l} \left[z^S_{k,l} - \tau_S + \eta_S \right]_+^2. \end{aligned} \tag{S35}$$

Here $[x]_+ = \max(x, 0)$, and η_S, η_C are safety margins that force the responses to stay away from the decision thresholds.

To avoid overfitting and maintain physical interpretability, the kernel coefficients are constrained to be bounded half-integers:

$$K_{ij}^{S}, K_{ij}^{C} \in \left\{ \frac{m}{2} \mid m \in \mathbb{Z} \right\}, \tag{S36}$$

$$-\kappa \leq K_{ij}^{S}, K_{ij}^{C} \leq \kappa. \tag{S37}$$

This constraint prevents the diagnostic head from becoming a black-box classifier and ensures that each coefficient has a clear physical meaning.

To account for occasional disturbances, a bounded perturbation term is also considered:

$$\|\Delta \mathbf{W}\|_{\infty} \leq \epsilon. \tag{S38}$$

For non-target responses, robustness requires

$$\max_{\|\Delta \mathbf{W}\|_{\infty} \leq \epsilon} \langle \mathbf{W} + \Delta \mathbf{W}, \mathbf{K} \rangle \leq \gamma - \eta. \tag{S39}$$

Equivalently,

$$\langle \mathbf{W}, \mathbf{K} \rangle + \epsilon \|\mathbf{K}\|_1 \leq \gamma - \eta. \tag{S40}$$

For target weak faults, robustness requires

$$\min_{\|\Delta \mathbf{W}\|_{\infty} \leq \epsilon} \langle \mathbf{W} + \Delta \mathbf{W}, \mathbf{K} \rangle \geq \gamma + \eta, \tag{S41}$$

which is equivalent to

$$\langle \mathbf{W}, \mathbf{K} \rangle - \epsilon \|\mathbf{K}\|_1 \geq \gamma + \eta. \tag{S42}$$

The practical optimization procedure is summarized as follows:

1. Construct normal, sensor-fault, and component-fault residual-pattern datasets according to the physical propagation rules of the EHSS.
2. Apply saturation and threshold normalization to both $\widetilde{\boldsymbol{D}}_{\nu,k}^{(A)}$ and $\widetilde{\boldsymbol{D}}_{\nu,k}^{(B)}$.
3. Extract four local windows $\mathbf{W}_{k,l}^{(b)}$ in each pressure branch.
4. Convolve both branches with the shared kernels, then fuse the responses into the five monitored positions $[i, x_s, P_A, P_B, x_c]^{\mathrm{T}}$.
5. Optimize $\boldsymbol{K}_S$ and $\boldsymbol{K}_C$ under classification, margin, robustness, bounded-weight, and half-integer constraints.
6. Select the kernel pair that satisfies all diagnostic inequalities while preserving physical interpretability.
7. Validate the selected kernels on normal, sensor-fault, and component-fault residual datasets.

Following the above optimization process, the final diagnostic kernels are selected as

$$\boldsymbol{K}_S = \begin{bmatrix} -0.5 & -0.5 & 1 \\ 1 & 1 & 1 \\ 2 & -4 & 0 \end{bmatrix}, \qquad \boldsymbol{K}_C = \begin{bmatrix} -1 & 0 & 1 \\ 1 & 3 & -3 \\ -2 & 2 & 0 \end{bmatrix}. \tag{S43}$$

The boundary entries for the sensor and component branches are denoted by $\zeta_S = 1$ and $\zeta_C = 4$, respectively. These two constants fill missing inference entries at the boundaries of the hierarchy. Their values are determined by considering both the treatment of missing information and the fault-detection thresholds. In the sensor diagnostic branch, the missing entry is assigned a value of 1. Saturation maps this value to 0.8, which is subsequently removed by thresholding. It therefore contributes nothing to the sensor fault indicator, leaving the diagnosis dependent on the actual residuals.

In the component diagnostic branch, missing terminal information would otherwise leave the downstream propagation pattern incomplete. A fixed completion value is therefore introduced for boundary-pattern assessment. The selected value of 4 produces a baseline response of 4 after saturation, thresholding, and convolution-kernel weighting, remaining below the component diagnostic threshold of 5. This completion alone cannot trigger an alarm: the net weighted contribution from the actual residuals must exceed 1 for the component response to exceed the threshold. Boundary completion thus supports fault identification while retaining a margin between the baseline response and the alarm threshold.

These values are selected using predefined physical residual patterns and checked against the diagnostic thresholds, providing a feasible set of boundary-completion constants.

The validation results are summarized in Table S2. The selected kernels correctly diagnose all generated residual-pattern samples, including normal samples, sensor-fault samples, and component/cascade-fault samples.

Table S2: Validation results of the optimized diagnostic kernels under standard and disturbed residual-pattern conditions.

Condition	Number of samples	Correct diagnosis	Accuracy
Normal condition	200	200	100.0%
Sensor fault	400	400	100.0%
Component/cascade fault	500	500	100.0%
Standard validation total	1100	1100	100.0%
Noisy normal condition	2000	2000	100.0%
Sensor fault with random disturbance	2000	1831	91.55%
Component fault with random disturbance	2000	1972	98.60%

Therefore, $\boldsymbol{K}_S$ and $\boldsymbol{K}_C$ are not manually tuned CNN kernels. They are obtained through a physics-constrained optimization process based on EHSS residual topology. The positive weights enhance the target residual topology, while the negative weights suppress competing fault patterns. This design is suitable for zero-sample fault diagnosis because it does not require labeled target fault samples to train a fully learned diagnostic head. The resulting kernels maintain reliable diagnostic performance under random disturbances, demonstrating their robustness to noise in fault diagnosis as shown in the Table S2.

S2.2 Convolution-Based Multi-Level Information Fusion

Following the hierarchical signal construction and convolution-kernel design, the designed kernels are applied to extract and fuse diagnostic information across different physical levels. Specifically, the deviation matrices of the two pressure branches are processed using the sensor- and component-fault kernels to obtain local convolutional responses. These responses are then fused according to the physical hierarchy to construct diagnostic indicators for fault identification and localization.

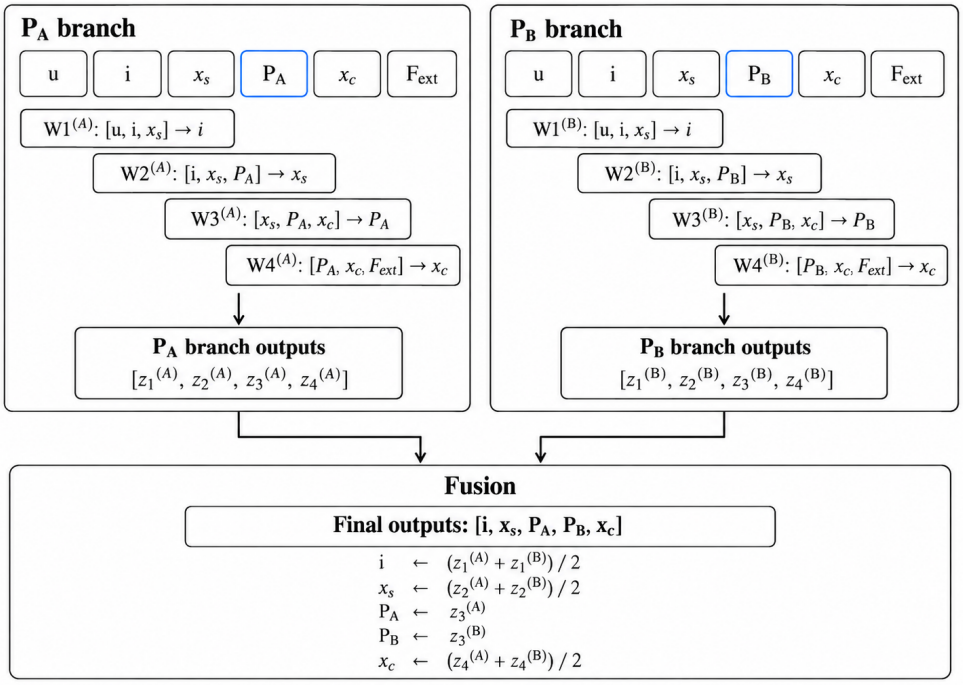


Fig. S2: Pressure-Branch Convolution and Indicator Fusion.

The chamber pressures are jointly propagated during hierarchical inference, with $\boldsymbol{P} = [P_A, P_B]^{\mathrm{T}}$. The normalized residuals of P_A and P_B are retained separately for convolutional feature extraction.

Let $\boldsymbol{d}_v \in \mathbb{R}^{1\times 3}$ denote the diagnostic matrix row associated with signal v. The internal rows contain the normalized upstream, PI-RNN, and downstream residuals, while the boundary rows follow Eq. (22) of the main text. The two pressure components are arranged into separate convolution branches, yielding

$$\boldsymbol{D}^{(A)} = \begin{bmatrix} \boldsymbol{d}_u \\ \boldsymbol{d}_i \\ \boldsymbol{d}_{x_s} \\ \boldsymbol{d}_{P_A} \\ \boldsymbol{d}_{x_c} \\ \boldsymbol{d}_{F_{\text{ext}}} \end{bmatrix}, \qquad \boldsymbol{D}^{(B)} = \begin{bmatrix} \boldsymbol{d}_u \\ \boldsymbol{d}_i \\ \boldsymbol{d}_{x_s} \\ \boldsymbol{d}_{P_B} \\ \boldsymbol{d}_{x_c} \\ \boldsymbol{d}_{F_{\text{ext}}} \end{bmatrix} \in \mathbb{R}^{6\times 3}. \tag{S44}$$

The two matrices share all rows except the pressure row. The shared residuals retain the coupled pressure information from the preceding hierarchical inference.

Following the preprocessing in Eq. (24) of the main text, each branch undergoes a 3×3 valid convo-

lution with unit stride and no padding:

$$\boldsymbol{z}^{(A)} = \mathrm{Conv}_{\mathrm{valid}}(\widetilde{\boldsymbol{D}}^{(A)}, \boldsymbol{K}) = \begin{bmatrix} z_1^{(A)} \\ z_2^{(A)} \\ z_3^{(A)} \\ z_4^{(A)} \end{bmatrix}, \qquad \boldsymbol{z}^{(A)}, \boldsymbol{z}^{(B)} \in \mathbb{R}^{4\times 1}, \tag{S45}$$

$$\boldsymbol{z}^{(B)} = \mathrm{Conv}_{\mathrm{valid}}(\widetilde{\boldsymbol{D}}^{(B)}, \boldsymbol{K}) = \begin{bmatrix} z_1^{(B)} \\ z_2^{(B)} \\ z_3^{(B)} \\ z_4^{(B)} \end{bmatrix},$$

where $\widetilde{\boldsymbol{D}}^{(A)}$ and $\widetilde{\boldsymbol{D}}^{(B)}$ denote the preprocessed matrices, and $\boldsymbol{K} \in \mathbb{R}^{3\times 3}$ is the corresponding diagnostic kernel.

The four convolution windows are $[u, i, x_s]$, $[i, x_s, P_A]$, $[x_s, P_A, x_c]$, and $[P_A, x_c, F_{\mathrm{ext}}]$ for branch A, and $[u, i, x_s]$, $[i, x_s, P_B]$, $[x_s, P_B, x_c]$, and $[P_B, x_c, F_{\mathrm{ext}}]$ for branch B. They produce four responses corresponding to i, x_s, P_A (or P_B), and x_c, respectively.

For each diagnostic type $\nu \in \{S, C\}$, the two branch outputs are fused into

$$\boldsymbol{\mathcal{I}}_\nu = \begin{bmatrix} (z_1^{(A)} + z_1^{(B)})/2 \\ (z_2^{(A)} + z_2^{(B)})/2 \\ z_3^{(A)} \\ z_3^{(B)} \\ (z_4^{(A)} + z_4^{(B)})/2 \end{bmatrix} \in \mathbb{R}^{5\times 1}. \tag{S46}$$

The five entries correspond to $[i, x_s, P_A, P_B, x_c]^{\mathrm{T}}$. The shared-channel responses are averaged, whereas the two pressure responses are retained separately to preserve chamber-specific diagnostic information.

S2.3 Fault localization

Based on the multi-level diagnostic indicators obtained through convolutional information fusion, fault localization is performed by analyzing their activation patterns along the physical hierarchy. Sensor faults are identified by localized inconsistencies, whereas component faults are characterized by deviations propagating to downstream levels. By combining these distinct patterns with the hierarchical dependencies of the EHSS, the proposed method determines the fault category and identifies its corresponding physical location. The monitored hierarchy is coil current i, spool displacement x_s, the paired chamber pressures (P_A, P_B), and piston displacement x_c. The two chamber pressures are at the same level and are interpreted jointly.

Diagnosis proceeds along the physical causal chain of the EHSS: coil current i drives spool displacement x_s; spool motion regulates the flow and determines the chamber pressures P_A and P_B; and these pressures drive the piston displacement x_c. The two chamber pressures are parallel signals at the same hydraulic level. The sensor indicator $I_{S,k}$ identifies a local anomaly in an individual measurement channel. For example, if $I_{S,k}$ persistently indicates an anomaly at x_s, while $I_{C,k}$ shows no corresponding downstream propagation pattern, the inconsistency is localized to the spool-displacement measurement channel

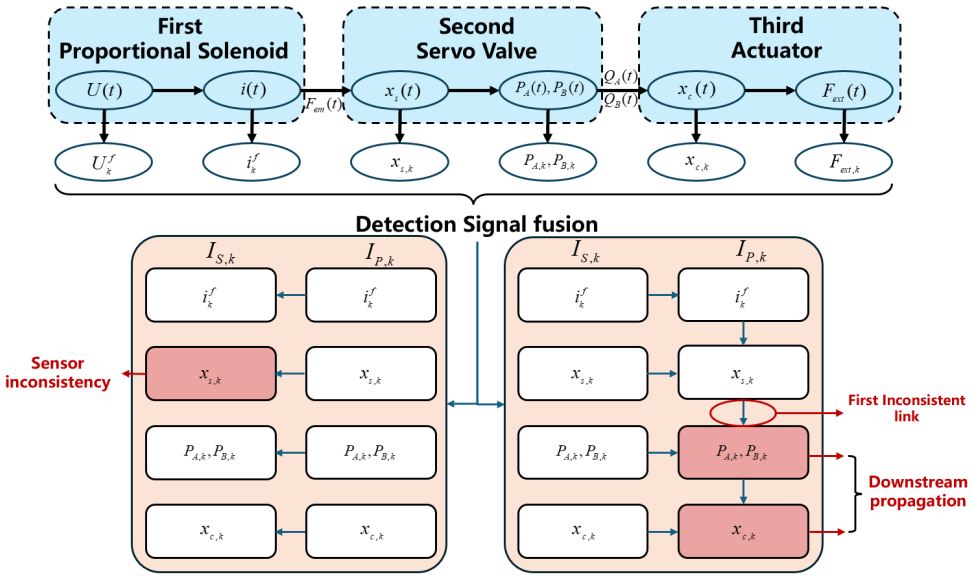


Fig. S3: Diagnostic logic for single-fault scenarios. The sensor indicator $\mathcal{I}_{S,k}$ identifies a local anomaly in an individual measurement channel, while the component indicator $\mathcal{I}_{C,k}$ identifies an inconsistency along the physical links in the causal chain.

and attributed to an LVDT sensor fault. By contrast, the component indicator $I_{C,k}$ is used to identify an inconsistency along the physical links in the causal chain. The signals are examined from upstream to downstream; when a level and the subsequent signals first deviate persistently from the expected pattern while the upstream signals remain consistent, the link between them is treated as the candidate fault location. In the spool-wear example, the component indicators for i and x_s remain normal, whereas the first inconsistency appears between x_s and the pressure level and then propagates to x_c. The diagnosis therefore focuses on the spool-displacement-to-flow/pressure conversion in the servo valve, consistent with spool wear, rather than on the LVDT or the hydraulic cylinder. A diagnosis is confirmed only when the relevant indicator exceeds its threshold for the required persistence duration, thereby excluding transient fluctuations.

Five-Seed Confusion Matrices and Healthy-Data False Alarms

The five seeds are 42, 123, 2024, 3407, and 5678. Each seed-specific matrix pools counts over 1, 5, 10, and 20 Hz. Per seed, there are 48,000 LVDT-fault segments after combining the two LVDT manifestations, 24,000 proportional-solenoid-fault segments, and 4,000 spool-fault segments, giving N_F = 76,000. The healthy evaluation contains 200 segments at each frequency, giving N_H = 800. These healthy segments were evaluated separately using each of the five trained models. Repeated predictions on the fixed test set do not constitute additional independent acquisitions.

Table S3: Healthy-data evaluation. All five seed-specific models give the counts shown at each frequency.

Excitation	Healthy segments	False alarms	FAR_H (%)	Specificity (%)
1 Hz	200	0	0.00	100.00
5 Hz	200	0	0.00	100.00

10 Hz	200	0	0.00	100.00
20 Hz	200	0	0.00	100.00
All frequencies	800	0	0.00	100.00

Table S4: PI-RNN diagnostic performance calculated from the five seed-specific confusion matrices. All rates are percentages; SD is the sample standard deviation across the five seeds.

Seed	Healthy FAR	Detection rate	Correct diagnosis	Missed detection	Unlocalized alarm
42	0.00	95.54	92.44	4.46	3.09
123	0.00	95.41	92.48	4.59	2.93
2024	0.00	96.10	92.49	3.90	3.61
3407	0.00	96.28	92.49	3.72	3.79
5678	0.00	95.72	92.49	4.28	3.23
Mean ± SD	0.00 ± 0.00	95.81 ± 0.37	92.48 ± 0.02	4.19 ± 0.37	3.33 ± 0.36

No false alarms were observed among the 800 healthy segments for any of the five models (Table S3). As shown in Table S4, the proposed method achieves a mean detection rate of 95.81 ± 0.37% and a fault-only correct-diagnosis rate of 92.48±0.02%, with 4.19±0.37% missed detections and 3.33±0.36% detected but unlocalized anomalies. The correct-localization rates are 95.09 ± 0.01% for LVDT faults, 87.58 ± 0.06% for proportional solenoid degradation, and 90.60 ± 0.13% for spool wear. No inter-class misclassifications were observed among the supported fault origins. These results are based on fault-only accuracy, with standard deviations reflecting training-seed variability on fixed test data.

The absence of observed healthy false alarms and inter-class misclassifications is consistent with the physics-guided diagnostic criteria, which require fault-specific indicator patterns to persist beyond the prescribed thresholds. However, physical operating limits may suppress certain residual responses and prevent complete fault localization. For example, when the actuator reaches its stroke limit, the piston-displacement residual may remain negligible despite a component fault. Consequently, spool wear may activate the pressure indicators without activating the piston-displacement indicator, resulting in an Other/Unlocalized diagnosis rather than an incorrect fault assignment.

Unlike conventional score-based classifiers, the proposed method determines fault origins using predefined physical fault signatures, incorporating hierarchical propagation patterns, fixed thresholds, and temporal persistence. Varying these thresholds may simultaneously affect fault detection, localization, and rejection decisions, rather than simply adjusting the sensitivity of a single classification score. Therefore, a conventional single-threshold ROC curve does not directly characterize the proposed structured diagnostic rule. Instead, performance is evaluated under the prescribed decision criteria, with an observed healthy FAR of 0% and a mean detection rate of 95.81%.

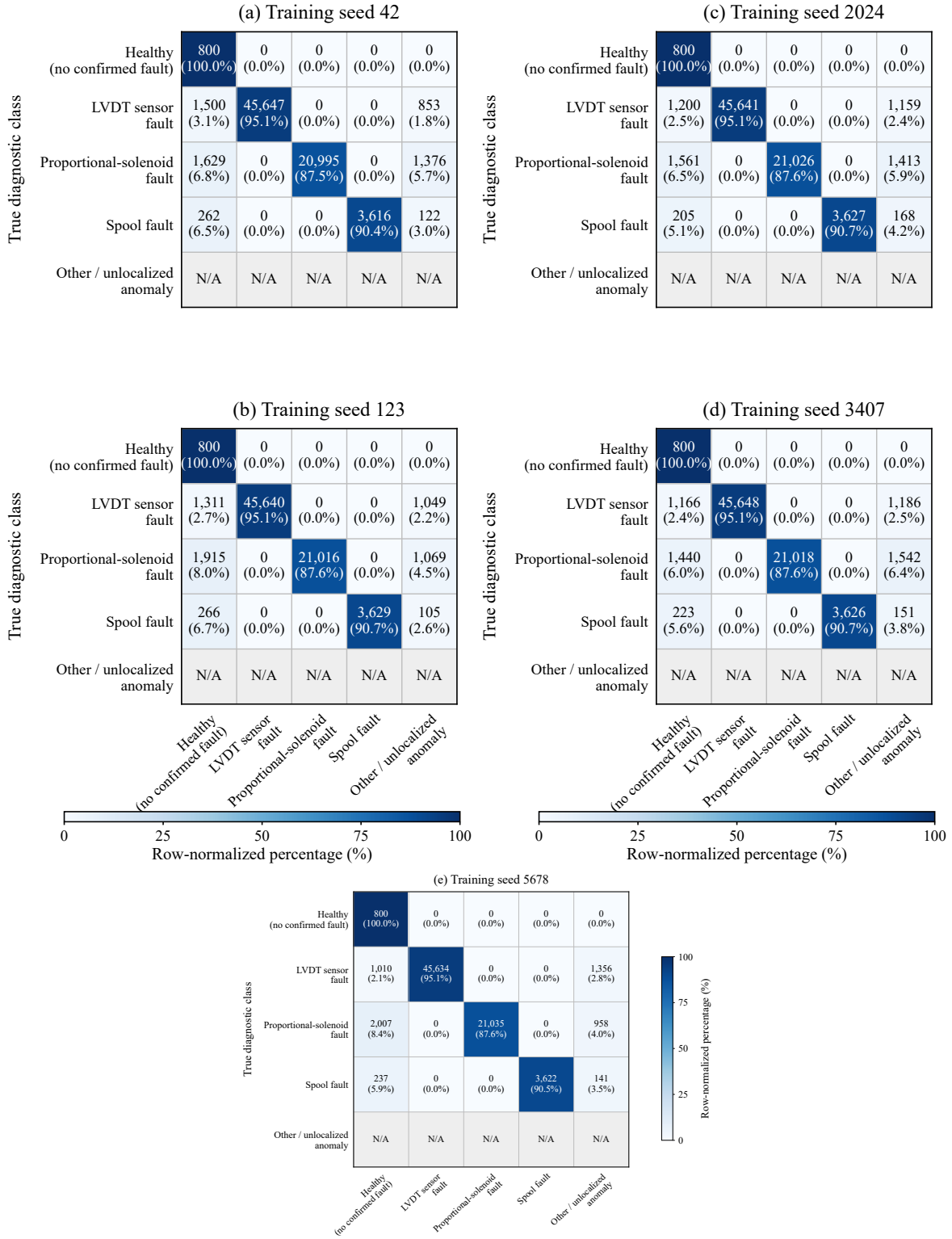


Fig. S4: PI-RNN confusion matrices for training seeds 42, 123, 2024, 3407, and 5678, pooling counts over the four excitation frequencies. The rows show the complete true diagnostic classes and the columns show the complete predicted diagnostic outcomes.

S3 Comparative Analysis of PI-RNN and EKF

S3.1 Implementation of the EKF.

To further evaluate the effectiveness of the proposed PI-RNN model, a comparison with the extended Kalman filter (EKF) is conducted using PI-RNN one-step predictions and EKF posterior state estimates. The EKF estimates system states through model-based prediction and recursive measurement correction, whereas the proposed PI-RNN incorporates RK4-based state propagation and trainable state-dependent dynamics to capture the normal behavior of the EHSS. This comparison examines their prediction and posterior-estimation accuracy and computational efficiency, highlighting the advantages of the proposed PI-RNN for residual-based fault diagnosis. The process and measurement-noise covariance matrices are denoted by $\mathbf{\Sigma}_w$ and $\mathbf{\Sigma}_v$, respectively, to distinguish them from hydraulic flow Q_A, Q_B. The state/output orders are

$$\mathbf{x}_{k+1} = f_d(\mathbf{x}_k, u_k) + \mathbf{w}_k, \qquad \mathbf{y}_k = \mathbf{H}\mathbf{x}_k + \mathbf{v}_k, \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_w), \quad \mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_v). \tag{S47}$$

To maintain consistency with the physical variables used in the PI-RNN, the EKF state vector is defined as

$$\mathbf{x}_k = [i_k,\ x_{s,k},\ \dot{x}_{s,k},\ P_{A,k},\ P_{B,k},\ Q_{A,k},\ Q_{B,k},\ x_{c,k},\ \dot{x}_{c,k}]^{\mathrm{T}},$$

where i is the proportional-solenoid current, x_s and $\dot{x}_s$ are the spool displacement and velocity, P_A and P_B are the chamber pressures, Q_A and Q_B are the corresponding flow rates, and x_c and $\dot{x}_c$ are the actuator displacement and velocity.

The measured-output vector is

$$\mathbf{y}_k = [i_k,\ x_{s,k},\ P_{A,k},\ P_{B,k},\ x_{c,k}]^{\mathrm{T}}.$$

Therefore, because these quantities are directly selected from the state vector, the measurement model is

$$\mathbf{y}_k = \mathbf{H}\mathbf{x}_k + \mathbf{v}_k,$$

with

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}. \tag{S48}$$

At each sampling instant, the EKF first performs a one-step prediction,

$$\hat{\mathbf{x}}_{k+1|k} = f_d(\hat{\mathbf{x}}_{k|k}, u_k),$$

and propagates the state covariance according to

$$\mathbf{\Sigma}_{k+1|k} = \mathbf{J}_k \mathbf{\Sigma}_{k|k} \mathbf{J}_k^{\mathrm{T}} + \mathbf{\Sigma}_w,$$

where the state-transition Jacobian is defined as

$$\mathbf{J}_k = \left. \frac{\partial f_d(\mathbf{x}, u_k)}{\partial \mathbf{x}} \right|_{\mathbf{x}=\hat{\mathbf{x}}_{k|k}}.$$

Because the EHSS model contains strongly nonlinear electromagnetic, mechanical, hydraulic-flow, and pressure dynamics, the Jacobian is evaluated numerically rather than analytically. Specifically, the implementation uses a central finite-difference approximation,

$$[\mathbf{J}_k]_{:,j} = \frac{f_d(\hat{\mathbf{x}}_{k|k} + \delta_j \mathbf{e}_j, u_k) - f_d(\hat{\mathbf{x}}_{k|k} - \delta_j \mathbf{e}_j, u_k)}{2\delta_j},$$

where $\mathbf{e}_j$ denotes the j-th unit vector and δ_j is the finite-difference perturbation for the corresponding state. Neither an analytical Jacobian nor automatic differentiation is used in the EKF implementation.

Before the measurement at $k+1$ is incorporated, the corresponding one-step predicted measurement is

$$\hat{\mathbf{y}}_{k+1|k} = \mathbf{H}\hat{\mathbf{x}}_{k+1|k}.$$

Once the new measurement $\mathbf{y}_{k+1}$ becomes available, the innovation is

$$\mathbf{r}_{k+1} = \mathbf{y}_{k+1} - \hat{\mathbf{y}}_{k+1|k},$$

with innovation covariance

$$\mathbf{\Sigma}^{\text{inn}}_{k+1} = \mathbf{H}\mathbf{\Sigma}_{k+1|k}\mathbf{H}^{\mathrm{T}} + \mathbf{\Sigma}_v.$$

The Kalman gain is then calculated as

$$\mathbf{G}_{k+1} = \mathbf{\Sigma}_{k+1|k}\mathbf{H}^{\mathrm{T}} \left(\mathbf{\Sigma}^{\text{inn}}_{k+1}\right)^{-1},$$

and the posterior state estimate is updated according to

$$\hat{\mathbf{x}}_{k+1|k+1} = \hat{\mathbf{x}}_{k+1|k} + \mathbf{G}_{k+1}\left(\mathbf{y}_{k+1} - \mathbf{H}\hat{\mathbf{x}}_{k+1|k}\right).$$

The posterior covariance is

$$\mathbf{\Sigma}_{k+1|k+1} = \left(\mathbf{I} - \mathbf{G}_{k+1}\mathbf{H}\right)\mathbf{\Sigma}_{k+1|k}.$$

Reorganizing the state-update equation gives

$$\hat{\mathbf{x}}_{k+1|k+1} = \underbrace{\left(\mathbf{I} - \mathbf{G}_{k+1}\mathbf{H}\right)}_{\mathbf{W}^{\text{pred}}_{k+1}}\hat{\mathbf{x}}_{k+1|k} + \underbrace{\mathbf{G}_{k+1}}_{\mathbf{W}^{\text{meas}}_{k+1}}\mathbf{y}_{k+1}.$$

Thus,

$$\mathbf{W}^{\text{pred}}_{k+1} = \mathbf{I} - \mathbf{G}_{k+1}\mathbf{H}$$

represents the contribution of the model-based prior estimate, whereas

$$\mathbf{W}^{\text{meas}}_{k+1} = \mathbf{G}_{k+1}$$

determines how the new sensor measurements modify that estimate.

For a directly measured scalar state, the update reduces to

$$\hat{x}^{+}_{k+1} = (1 - G_{k+1})\hat{x}^{-}_{k+1} + G_{k+1}y_{k+1}.$$

Accordingly,

$$w_{k+1}^{\text{pred}} = 1 - G_{k+1}, \qquad w_{k+1}^{\text{meas}} = G_{k+1}.$$

The diagonal covariance settings in the original SI coordinates (not the display units of Table S5) are

$$\begin{aligned}
\operatorname{diag}(\boldsymbol{\Sigma}_0) &= (2.5{\times}10^{-3}, 4{\times}10^{-10}, 10^{-4}, 4{\times}10^{10}, 4{\times}10^{10}, \\
&\qquad 10^{-8}, 10^{-8}, 4{\times}10^{-8}, 10^{-4}), \\
\operatorname{diag}(\boldsymbol{\Sigma}_w) &= (4{\times}10^{-4}, 4{\times}10^{-12}, 2.5{\times}10^{-5}, 2.5{\times}10^{9}, 2.5{\times}10^{9}, \\
&\qquad 10^{-10}, 10^{-10}, 10^{-10}, 4{\times}10^{-6}), \\
\operatorname{diag}(\boldsymbol{\Sigma}_v) &= (9.38{\times}10^{-3}, 1.05{\times}10^{-9}, 1.72{\times}10^{10}, 2.10{\times}10^{11}, 8.21{\times}10^{-8}).
\end{aligned} \tag{S49}$$

In the measured-output order $[i, x_s, P_A, P_B, x_c]$, the corresponding measurement-noise standard deviations in display units are approximately 0.097 A, 0.032 mm, 1.31 bar, 4.58 bar, and 0.287 mm.

S3.2 Integration of Dynamic L_f into the EKF.

The dynamic-L_f EKF retains the state vector, observation matrix, and prediction–correction recursion of Section S3.1. The change is confined to the flow-state transition. For each chamber $j \in \{A, B\}$, the fixed inertance in the flow equation is replaced by a positive state-dependent value:

$$\dot{Q}_j = \frac{f_j(x_s, P_A, P_B) - R_{f,j} Q_j}{L_{f,j}^{\text{EKF}}(\mathbf{x}, t)}, \qquad L_{f,j}^{\text{EKF}}(\mathbf{x}, t) = L_{f,\min} + \text{Softplus}\left[\mathcal{N}_{\boldsymbol{\theta}_L}\left(\boldsymbol{P}, x_s, x_c, \dot{x}_s, \dot{x}_c\right)\right] \tag{S50}$$

Here f_j is the corresponding valve-flow term already defined by the physical model; no additional correction is applied to the pressure, spool, or actuator equations. The mapping parameters θ are identified using healthy data and frozen during EKF evaluation. The filter evaluates $L_{f,A}^{\text{EKF}}$ and $L_{f,B}^{\text{EKF}}$ at the current propagated state rather than augmenting $\mathbf{x}$ with online parameter estimates. For the 20-Hz implementation, $g_\theta(\mathbf{x}, t) = \mathbf{W}_L \mathbf{z} + \mathbf{b}_L$, where $\mathbf{z} = [P_A, P_B, x_s, x_c, \dot{x}_s, \dot{x}_c]^{\mathrm{T}}$; The archived 1-Hz EKF run used a previously fitted 14-coefficient state/phase basis. It is retained as a separate numerical comparison and is not presented as an evaluation of the final six-input affine mapping.

The state-dependent transition f_d^{dyn} is obtained by numerically integrating Eq. (S50) with the other state equations. Its Jacobian is computed by the same central-difference procedure as in Section S3.1, so each perturbed transition reevaluates the dynamic L_f. The predicted covariance and the innovation, gain, state correction, and posterior covariance then follow the EKF equations in that section. The fixed-L_f control replaces $L_{f,j}^{\text{EKF}}$ by a constant in the same flow equation.

S3.3 PI-RNN and EKF Results at 1 and 20 Hz.

The PI-RNN output is its one-step state prediction. The EKF outputs are posterior estimates after the current measurement is assimilated at both frequencies.

For each measured channel j, the reported relative error is

$$E_{j,L_2}(\%) = 100\, \frac{\|\widehat{\mathbf{y}}_j - \mathbf{y}_j^f\|_2}{\|\mathbf{y}_j^f\|_2}, \tag{S51}$$

where $\mathbf{y}_j^f$ is the filtered measurement over the indicated evaluation points and $\widehat{\mathbf{y}}_j$ is the displayed method output.

(1)1-Hz comparison

Table S5 and Figs. S5–S6 reproduce the three-method comparison: PI-RNN prediction, the originally reported fixed-L_f EKF posterior estimate, and the dynamic-L_f EKF posterior estimate. The dynamic-L_f configuration has relative L_2 errors of 5.761%, 1.649%, 2.034%, 0.934%, and 4.884% for i, x_s, x_c, P_A, and P_B, respectively.

Table S5: Relative L_2 errors (%) for the 1-Hz record. The PI-RNN column is a one-step prediction; both EKF columns are posterior estimates.

State	PI-RNN	Fixed-L_f EKF	Dynamic-L_f EKF
i	1.898	5.992	5.761
x_s	1.501	1.655	1.649
x_c	1.086	6.895	2.034
P_A	1.511	0.936	0.934
P_B	3.098	4.893	4.884
Mean	1.819	4.074	3.052

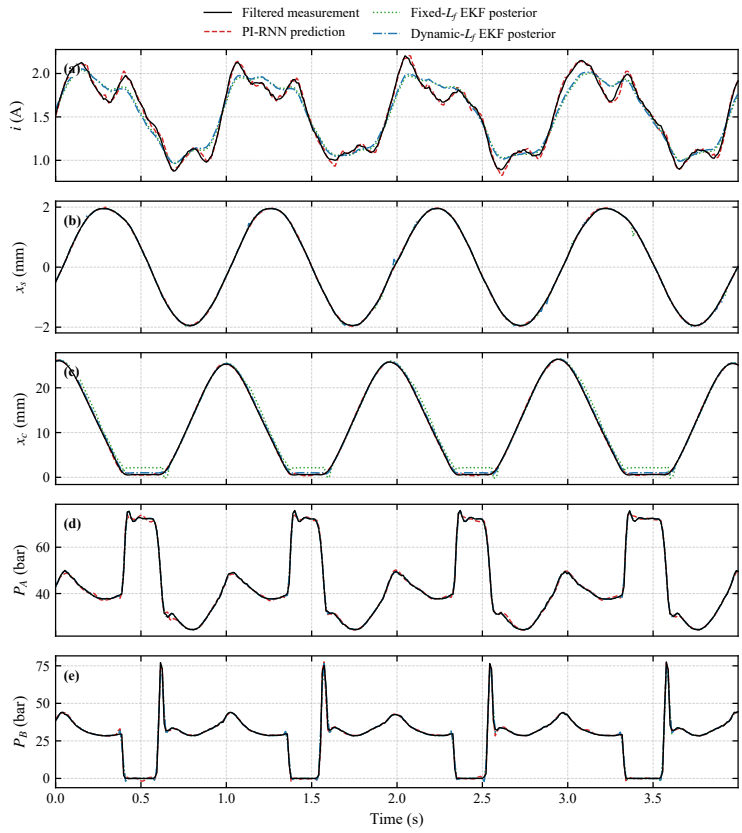


Fig. S5: Filtered measurements, PI-RNN one-step prediction, fixed-L_f EKF and dynamic-L_f EKF posterior estimates for the 1-Hz record.

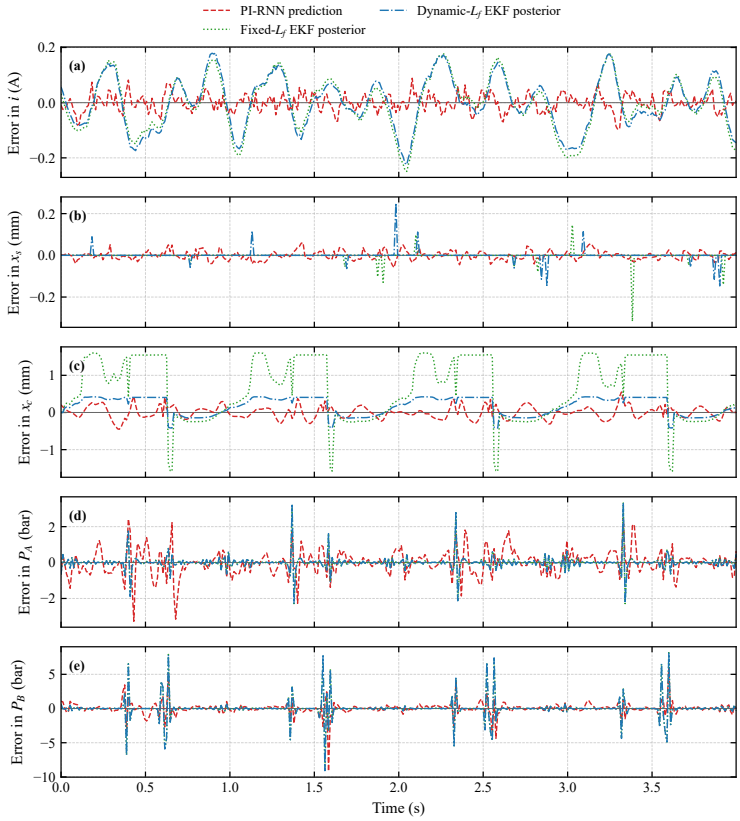


Fig. S6: PI-RNN one-step prediction errors and EKF posterior-estimation errors relative to filtered measurements for the three outputs in Fig. S5.

(2) 20-Hz comparison

At 20 Hz, Fig. S7 compares PI-RNN one-step predictions and EKF posterior estimates with the ground-truth measurements, while Fig. S8 presents the corresponding prediction and posterior-estimation errors. Table S6 summarizes the average prediction and posterior-estimation errors over the evaluation interval.

Table S6: Relative L_2 errors (%) for the 20-Hz record. The PI-RNN column is a one-step prediction; both EKF columns are posterior estimates.

State	PI-RNN	Fixed-L_f EKF	Dynamic-L_f EKF
i	0.800	2.098	1.291
x_s	3.051	8.274	7.385
x_c	7.020	10.350	10.271
P_A	2.761	3.506	3.231
P_B	14.254	20.851	20.164
Mean	5.577	9.016	8.469

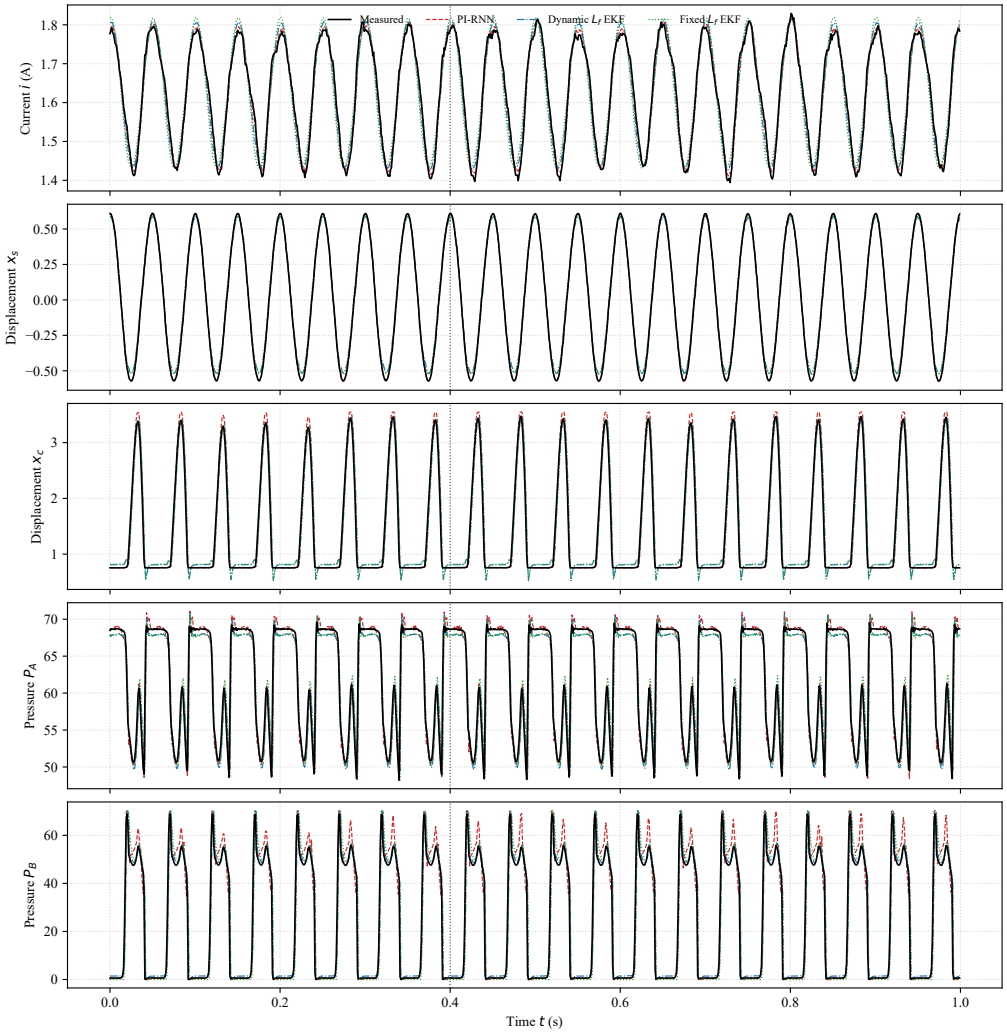


Fig. S7: PI-RNN one-step predictions and fixed-L_f and dynamic-L_f EKF posterior estimates on the 20-Hz record.

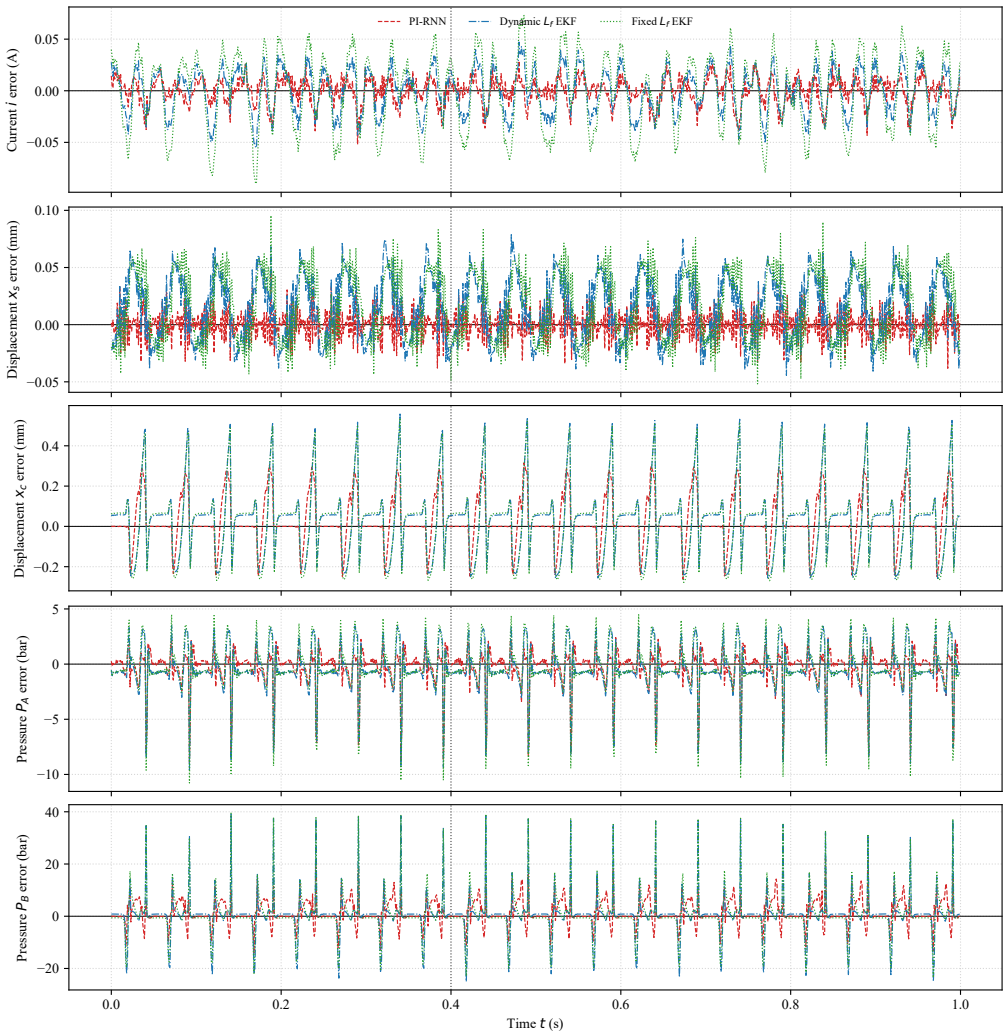


Fig. S8: PI-RNN one-step prediction errors and EKF posterior-estimation errors relative to filtered measurements for the three outputs in Fig. S7.

At 1 Hz, the hybrid PI-RNN achieves a lower mean relative L_2 error (1.819%) than the fixed-L_f EKF (4.074%) and dynamic-L_f EKF (3.052%), although both EKFs perform slightly better for P_A.

At 20 Hz, PI-RNN is evaluated using one-step predictions, while both EKFs are evaluated using posterior estimates. The hybrid PI-RNN achieves lower errors across all five monitored signals, with a mean relative L_2 error of 5.577%, compared with 9.016% and 8.469% for the fixed-L_f and dynamic-L_f EKFs, respectively.

Overall, these results demonstrate that the proposed RK4-integrated hybrid PI-RNN achieves lower mean errors in the comparisons with the evaluated EKF posterior estimates.

S4 Learnability and Identifiability of L_f

S4.1 1. Convergence and practical identifiability of L_f

In the proposed PI-RNN model, the hydraulic inertance parameters $L_{f,A}$ and $L_{f,B}$ are not directly measured in the experiment. However, they explicitly appear in the valve flow dynamic residuals and are constrained by the coupled pressure dynamics, reconstructed chamber flow rates, spool displacement, and actuator motion. Therefore, these parameters can be estimated indirectly by minimizing the discrepancy between the measured system response and the physics-constrained PI-RNN prediction.

To examine the effect of these trainable parameters, we first tested the case where L_f was fixed. As shown in Fig. S9, the normalized loss remained relatively large, indicating that a fixed inertance parameter was insufficient to represent the unsteady flow dynamics under the tested operating conditions.

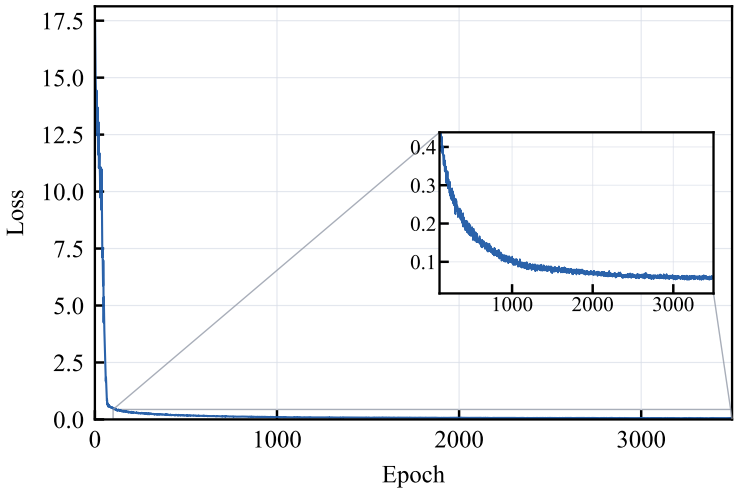


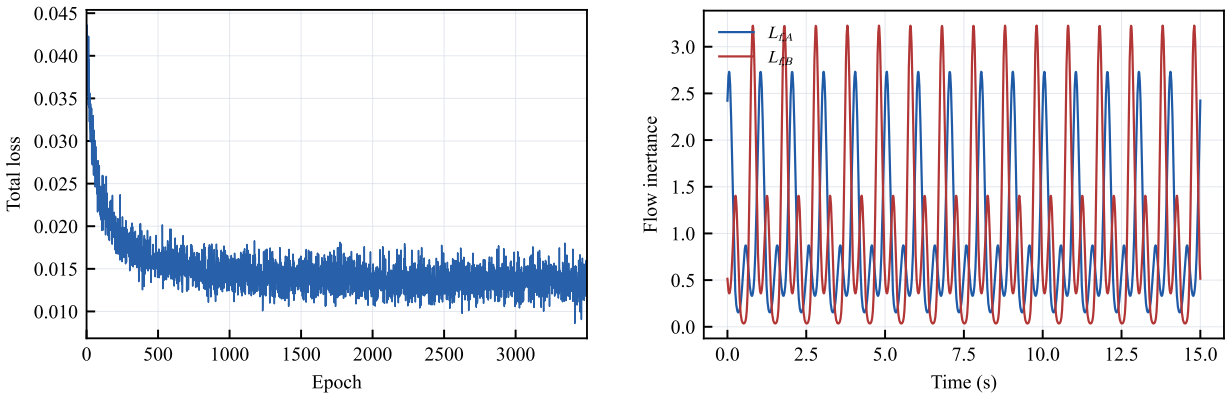
Fig. S9: Overall loss curve with fixed flow inertance parameter.

We then treated the equivalent flow inertance as a trainable state-dependent mapping in the PI-RNN framework. The mapping is the same six-input, two-output affine map followed by componentwise Softplus as in the manuscript:

$$\begin{aligned} \mathbf{z} &= [P_B, P_A, x_s, x_c, \dot{x}_s, \dot{x}_c]^{\mathrm{T}}, \\ \mathbf{L}_f(\mathbf{z}) &= L_{f,\min} + \boldsymbol{T}_j \,\mathrm{Softplus}(\mathbf{W}_L \mathbf{z} + \mathbf{b}_L), \end{aligned} \tag{S52}$$

where $\mathbf{W}_L \in \mathbb{R}^{2\times 6}$ and $\mathbf{b}_L \in \mathbb{R}^2$ contain 14 trainable scalar coefficients in total, and $L_{f,\min} > 0$ imposes a positive lower bound. The mapping parameters are fixed after training, while the evaluated $\mathbf{L}_f$ varies with the instantaneous state.

After retraining L_f as a variable parameter, the overall loss was reduced by approximately 85%, as shown in Fig. S10a. This result indicates that the hydraulic inertance term contributes meaningful information to the PI-RNN loss formulation and can be effectively estimated under the tested excitation conditions.



(a) Overall loss curve after retraining the equivalent flow inertance.

(b) Identified equivalent flow inertance $L_{f,A}(t)$ and $L_{f,B}(t)$.

Fig. S10: Retraining loss and identified equivalent flow-inertance trajectories.

The identified equivalent inertance trajectories are shown in Fig. S10b, and the corresponding trainable coefficient trajectories are shown in Fig. S11. These results show that the learned inertance terms remain bounded and improve the dynamic consistency of the PI-RNN prediction.

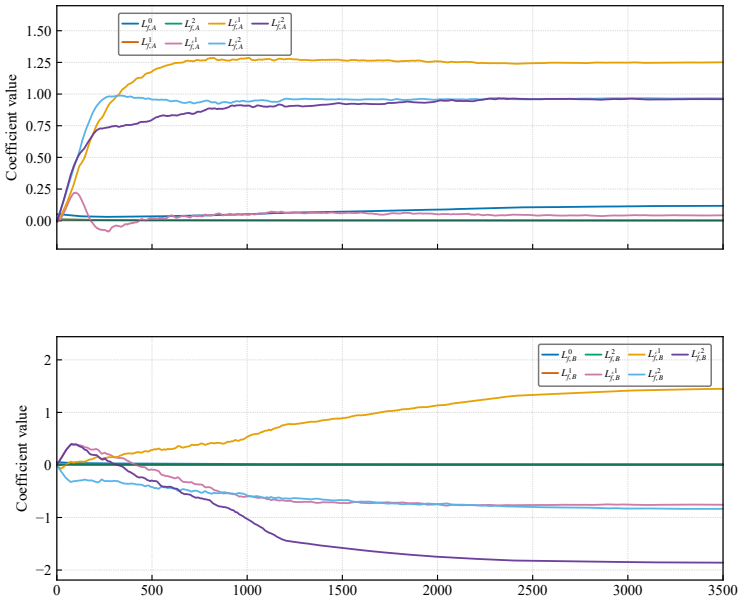


Fig. S11: Trainable coefficient trajectories for $L_{f,A}(t)$ and $L_{f,B}(t)$.

S4.2 Convergence Analysis of L_f Under Different Initializations

We initialize L_f from several initial values and repeat the complete training procedure. Despite the different initial conditions, the estimated L_f converges to nearly the same value/function. This indicates that the final estimate is not strongly dependent on the initial guess.

Fig. S12 shows the convergence trajectories of L_f under different initial values, demonstrating that all runs gradually approach a similar solution.

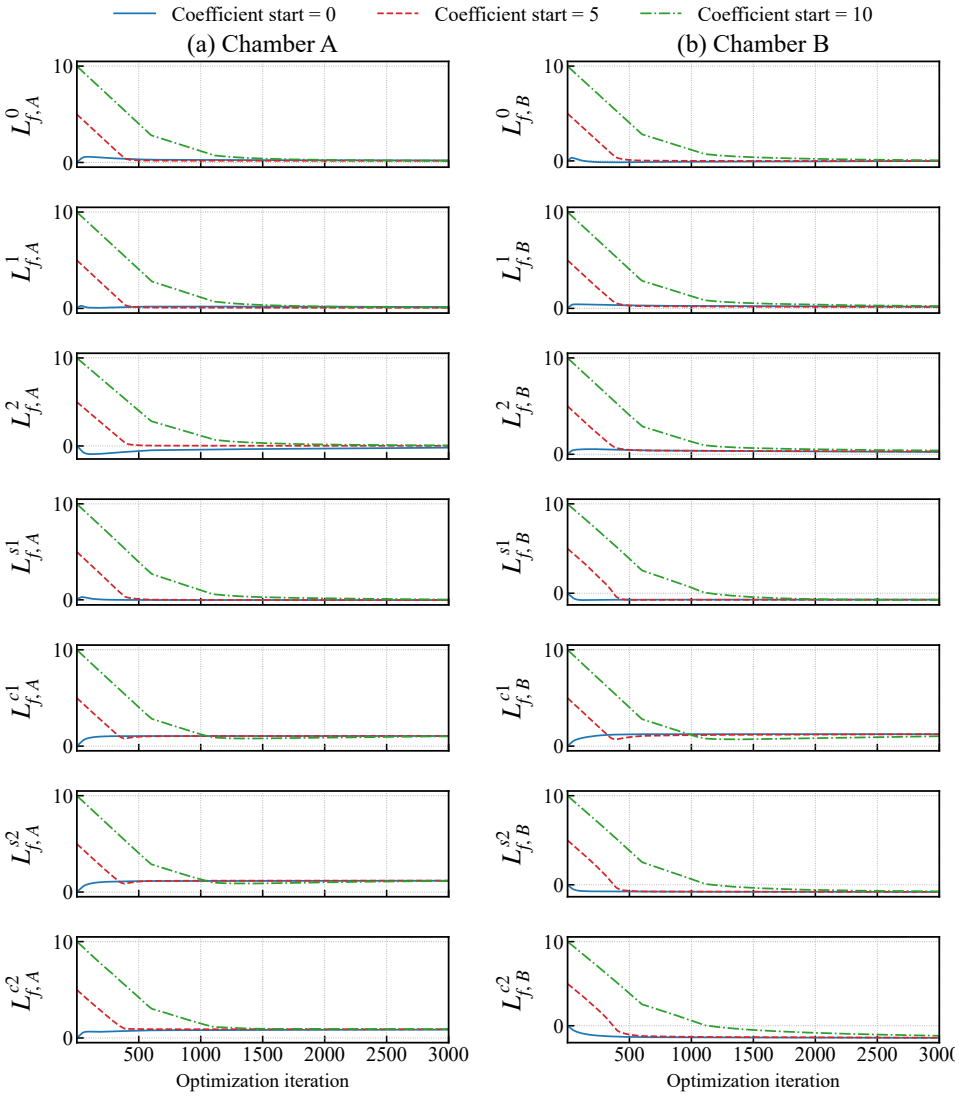


Fig. S12: Convergence trajectories of L_f under different initial values.

S4.3 Reasons for absorbing unmodeled physical quantities

The learned L_f contain a contribution from dynamics that are not explicitly included in the reduced-order flow model. For hydraulic path $j \in \{A, B\}$, the actual flow dynamics can be written as

$$L^0_{f,j}\dot{Q}_j + R_{f,j}Q_j = f_j(x_s, P_A, P_B, Q_j) + d_j, \tag{S53}$$

where $L^0_{f,j}$ is the nominal flow-inertance parameter and d_j denotes the neglected dynamics, such as local flow transients, pressure-dependent effects, and other higher-order hydraulic behavior.

In the proposed model, these dynamics are not represented by an additional neural residual. Instead, a state-dependent effective parameter is used:

$$L^{\text{eff}}_{f,j}(\boldsymbol{h})\dot{Q}_j + R_{f,j}Q_j = f_j(x_s, P_A, P_B, Q_j). \tag{S54}$$

Comparing the two equations gives

$$\left[L^{\text{eff}}_{f,j}(\boldsymbol{h}) - L^0_{f,j}\right]\dot{Q}_j = -d_j. \tag{S55}$$

This relation shows that model discrepancies that can be represented through the transient-flow term may appear as a change in the estimated L_f. Therefore, the learned quantity should be interpreted as a *state-dependent effective flow-inertance parameter*, rather than the nominal physical inertance alone.

The effective parameter is represented as

$$L^{\text{eff}}_{f,j}(\boldsymbol{h}) = L_{f,\min} + \boldsymbol{T}_j \,\text{Softplus}\left[\mathcal{N}_{\boldsymbol{\theta}_L,j}(x_s, \boldsymbol{P}, x_c, \dot{x}_s, \dot{x}_c)\right], \tag{S56}$$

which allows L_f to vary with the hydraulic state while maintaining $L^{\text{eff}}_{f,j} > 0$.

Thus, L_f can absorb part of the unmodeled dynamics, but this compensation is introduced through the prescribed $L_f\dot{Q}_j$ term rather than through an unrestricted additive residual. The present data do not allow the nominal physical contribution and the absorbed model discrepancy to be separated uniquely.

S5 Effects of physical-parameter variation on diagnostic indicators

This section compares 20-Hz healthy-data indicators with nominal L_f and with $0.90L_f$. Fig. S13 reproduces only the diagnostic-indicator panels of that artwork. The trajectories are omitted for compactness. Model mismatch changes the responses, particularly near the actuator-displacement stage, illustrating sensitivity to the flow-model parameter rather than immunity to parameter variation.

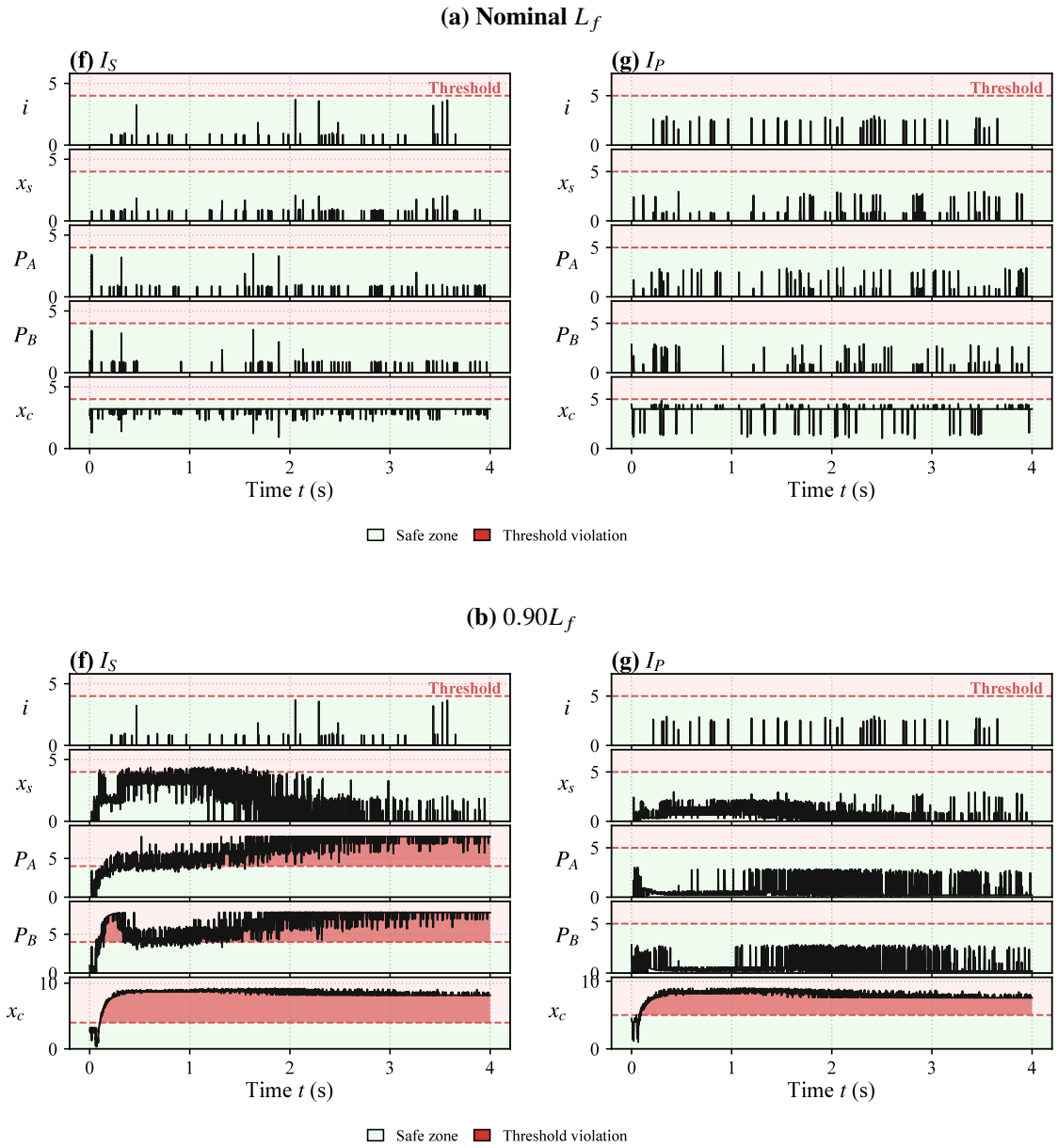


Fig. S13: Sensor indicators I_S (left) and component indicators I_C (right) under nominal and perturbed flow-inertance settings. Red dashed lines show the original thresholds. These are qualitative examples, not confirmed false-alarm-rate estimates.

An instantaneous threshold crossing is not a confirmed alarm: the $0.2T_c$ continuous-duration rule must be applied to the underlying records. The archived artwork provides no verified confirmed-alarm count or trial denominator, so a numerical healthy false-alarm rate is not inferred. Varying L_f also does not replace independent bulk-modulus and friction-variation tests. Those remain separate validation requirements.

S6 Transfer-learning baselines, CNN ablation, and EHSS comparison protocol

The two literature baselines are blind domain transition (BDT), introduced for motor-bearing health monitoring by Kiranyaz et al. [1], and the zero-shot transfer-learning method introduced for structural damage detection by Heravi et al. [2]. BDT uses a one-dimensional operational generative adversarial network (Op-GAN) to learn the transition from normal to faulty signals from source-machine data, synthesize potential faulty signals for a target machine, and train a target-domain detector using the target machine's healthy signals and the synthesized faulty signals. Heravi et al.'s autoencoder-based feature-mapping transfer-learning method (AFMTL), which maps healthy target-domain data to the source domain and applies a source-trained one-class support vector machine (OC-SVM) for damage detection, is referred to here as AFMTL-OCSVM. The CNN-based ablation replaces the hybrid PI-RNN predictor with a CNN while retaining the hierarchical multi-source diagnosis framework. Its implementation is described in Section S6.3.

The manuscript evaluates LVDT drift, LVDT sensitivity degradation, proportional-solenoid degradation, and spool wear at excitation frequencies of 1, 5, 10, and 20 Hz. The first three scenarios use severity settings of Mild(5%, 10%), Moderate(15%, 20%), and Severe(25%, 30%). Spool wear uses one worn-spool condition.

S6.1 BDT

S6.1.1 Original method.

BDT first learns a healthy-to-fault signal transformation from labeled healthy and faulty source-machine signals using a one-dimensional operational generative adversarial network (Op-GAN). Its generator then transforms healthy target-machine signals into synthetic faulty signals. A separate one-dimensional self-organized operational neural network (Self-ONN) is trained with real healthy and synthetic faulty target signals to make binary healthy/faulty decisions. Actual target faults are used for evaluation rather than detector training. Thus, the original method uses source-domain fault knowledge even though target-domain fault observations are unavailable during training [1].

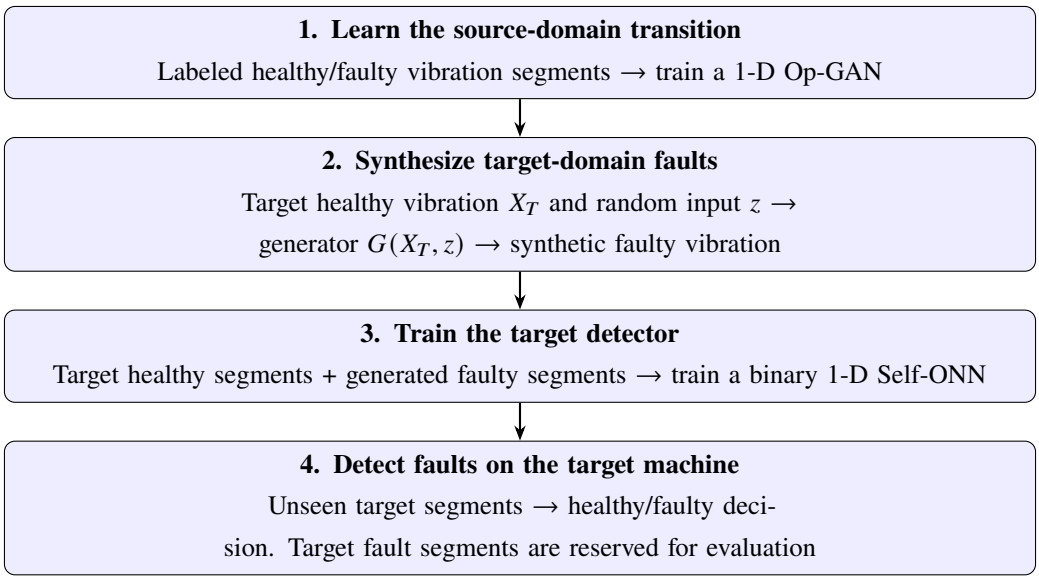


Fig. S14: Implementation flow of the original BDT method reported by Kiranyaz et al. [1].

Original method Step 1: Learn the healthy-to-fault transition The source-domain Op-GAN is trained with labeled healthy and faulty vibration segments. The generator receives a healthy segment X_S and a random input z and produces a fault-like segment,

$$\widehat{Y}_S = G(X_S, z). \tag{S57}$$

The discriminator provides the adversarial signal, while temporal and spectral reconstruction terms encourage the generated segment to reproduce fault-signal characteristics.

Original method Step 2: Generate target-domain fault samples After Op-GAN training, the generator is applied to healthy signals from a new target machine. The target healthy signal and a random input are used to produce a synthetic fault counterpart. This step transfers the learned source-domain healthy-to-fault transition to the target signal domain. Real target fault segments are not needed to train the generator for this application.

Original method Step 3: Train and evaluate the target detector The target detector is trained on the target machine's real healthy segments and the generated fault segments. The original study uses a lightweight Self-ONN for the binary healthy/faulty decision. Actual target fault data are held out until evaluation, so the detector is trained without target fault examples. Figure S14 summarizes this sequence.

In the published experiments, source-machine records from a held-out speed were reserved for validation. The target healthy training subset was used for fault synthesis and detector training. The held-out healthy data and the actual target fault data were used for testing.

The original Op-GAN combines a binary adversarial loss with temporal and spectral reconstruction losses. Its objective is

$$\mathcal{L}_{\text{Op-GAN}} = \mathcal{L}_{\text{BCE}} + \lambda \left(\mathcal{L}_{\text{time}} + \mathcal{L}_{\text{spectral}} \right). \tag{S58}$$

In the original study, the weighting parameter was set to $\lambda = 100$.

EHSS-adapted implementation.

In the EHSS-adapted BDT implementation, the source domain comprises simulated healthy and faulty EHSS responses under different excitation frequencies and fault types, whereas the target domain comprises measured EHSS samples. Only healthy target-domain samples are available during model adaptation and detector training. All target-domain faulty samples are strictly reserved for final evaluation.

To apply BDT to EHSS fault diagnosis, the original motor-bearing workflow was modified to accommodate EHSS data while retaining the underlying BDT principle. The following stages detail the adapted data processing and diagnostic procedures. The target detector is extended to five classes,

$$C_{\text{BDT}} = \{0 : \text{Healthy}, 1 : \text{LVDT drift}, 2 : \text{LVDT degradation}, 3 : \text{Solenoid degradation}, 4 : \text{Spool wear}\}. \tag{S59}$$

LVDT drift and degradation are treated as separate classes in BDT training, prediction, and confusion-matrix plotting. These multiclass extensions require new training and testing. The previous binary detection rates are not relabeled as multiclass results.

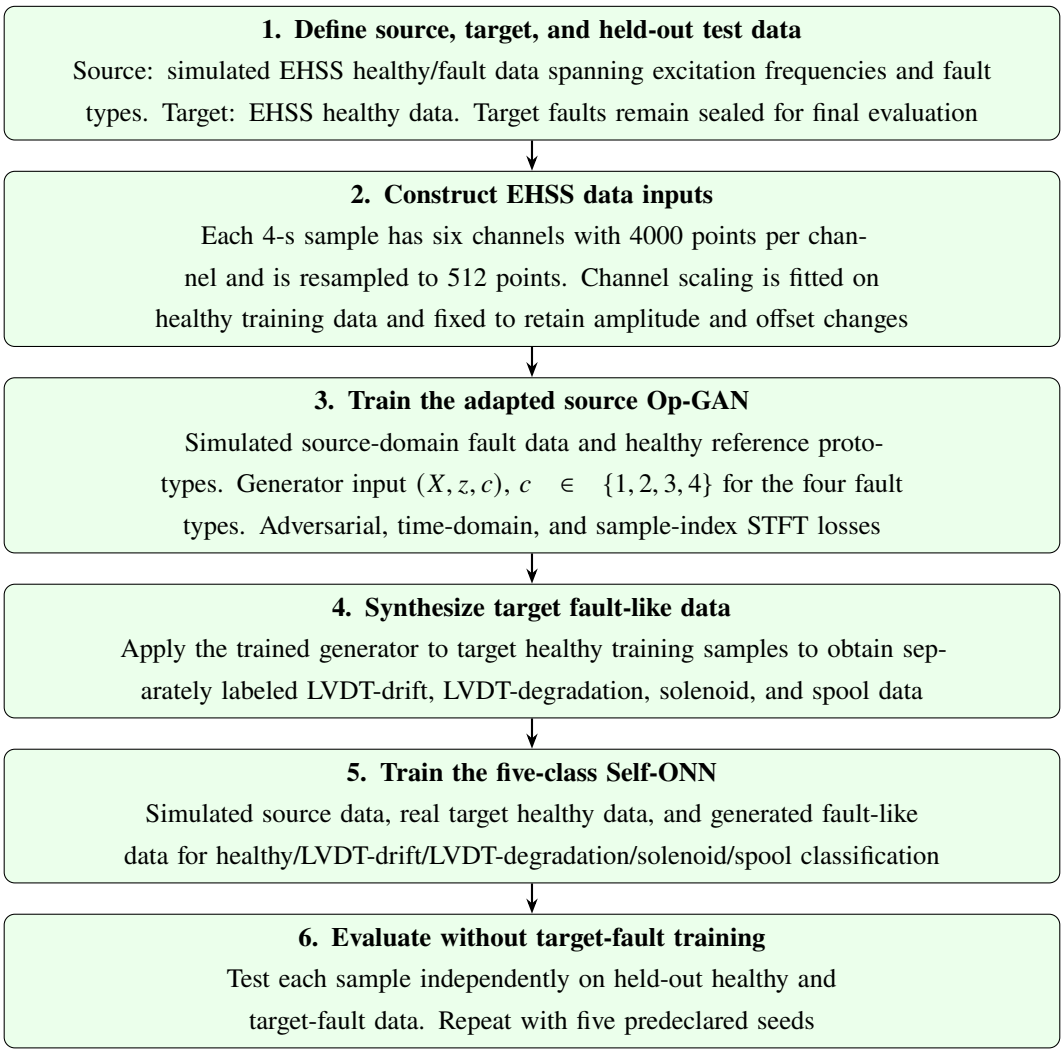


Fig. S15: EHSS-adapted BDT workflow. The numbered stages correspond to the implementation subsections below.

EHSS Step 1: Define domains and data partitions

The source domain consists of simulated healthy and labeled faulty data spanning different excitation frequencies and fault types, with labels defined in Eq. (S59). Fault severity is treated as within-class variation and is neither modeled nor identified. The source data are split into training and validation subsets, and the validation subset is used only for checkpoint selection. The target-domain healthy samples are split into training and test subsets. The target training subset is used for scaling, reference estimation, fault synthesis, and detector training, and the target test subset is used only for final evaluation. Actual target fault samples are excluded from training, validation, scaling, and reference estimation, and are used only in the final test. The 4-s segments are non-overlapping, and all partitions are identical across the five seeds.

EHSS Step 2: Construct and preprocess sample inputs

Each sample has six channels: input voltage u, drive current i, spool displacement x_s, actuator displacement x_c, and chamber pressures P_A and P_B. Each sample contains 4 s of data acquired at 1000 Hz, giving 4000 points per channel. Before being fed into the network, each sample is resampled with anti-aliasing filtering to 512 points

per channel. This corresponds to an effective rate of 128 Hz, so the retained band is limited to 0–64 Hz, which covers the 1–20 Hz excitation frequencies studied here. Channel-wise min–max parameters are fitted once for each domain, using the source healthy training subset for the source domain and the target healthy training subset for the target domain, and are then fixed for all classes. Values outside the fitted range, such as those of fault samples, are not clipped. Unlike the per-segment min–max normalization of the original method, this fixed scaling preserves the amplitude and offset changes caused by LVDT gain and offset faults. Target fault samples are scaled with the fixed target parameters and are used only for final testing.

EHSS Step 3: Train the adapted source Op-GAN

The adapted Op-GAN is trained using simulated source-domain EHSS data, including healthy samples and labeled fault samples spanning different excitation frequencies and fault types. It retains the source method's third-order operational layers ($Q = 3$), U-Net skip connections, and joint adversarial, temporal, and spectral training objective. The operational mapping is implemented as a convolution of the concatenated input, its elementwise square, and its elementwise cube.

The generator is a 1-D U-Net with five encoder and five decoder layers. The encoder channel widths are $(32, 64, 128, 256, 512)$, and the decoder channel widths are $(256, 128, 64, 32, 6)$. All generator kernels have length 4, stride 2, and padding 1. Hidden layers use tanh and the final output is linear. The temporal length decreases from 512 to 16 points through the encoder and is restored to 512 points by the decoder. A four-component one-hot code $\mathbf{e}_c \in \mathbb{R}^4$ (the c-th unit vector) identifies the requested fault type and is broadcast along the sample. Different severity levels within a fault type share the same code. The generator input has 10 channels: six healthy signal channels and four fault-type code channels. In place of the random input z of the original Op-GAN, the healthy input is perturbed by $\boldsymbol{\delta} = \boldsymbol{\sigma}_\delta \odot \mathbf{z}$, where the entries of $\mathbf{z}$ are independently drawn from a standard normal distribution and $\boldsymbol{\sigma}_\delta$ is a per-channel scale, in the fixed scaled coordinates, estimated from the residuals of the target healthy training samples about their frequency-wise mean. The perturbation mimics the individual noise and fluctuation of measured signals. During generator training, the same $\boldsymbol{\delta}$ is also added to the supervising source fault sample, so that the generator learns the fault-induced residual instead of removing the perturbation:

$$\widehat{\mathbf{Y}}_S^c = G(\mathrm{cat}[\mathbf{X}_S + \boldsymbol{\delta}, \mathbf{e}_c]), \qquad \mathbf{Y}_S^c \leftarrow \mathbf{Y}_S^c + \boldsymbol{\delta}, \qquad c \in \{1, 2, 3, 4\}. \tag{S60}$$

The conditional discriminator receives a 16-channel concatenation. For a real pair, this consists of the perturbed healthy reference prototype $\mathbf{X}_S + \boldsymbol{\delta}$, the perturbed labeled source fault sample $\mathbf{Y}_S^c$, and the class code $\mathbf{e}_c$. For a generated pair, it consists of the same healthy input $\mathbf{X}_S + \boldsymbol{\delta}$, its generated fault-like counterpart $\widehat{\mathbf{Y}}_S^c$, and the same class code. The discriminator has five third-order operational layers ($Q = 3$) with kernel length 4, stride 2, and padding 1 and widths $(16, 32, 64, 64, 64)$, followed by a one-channel output layer with kernel length 3 and stride 1. The mean of the output patch logits is used as the scalar discriminator score.

At each update, the four fault types are sampled uniformly, with severity variation retained among the source samples within each type. Severity is not conditioned on or identified. The generator therefore learns a representative fault pattern for each fault type. Each simulated source fault sample is paired with the healthy mean reference prototype $\mathbf{X}_S$ at its excitation frequency, which is constructed only from simulated healthy training data. Real target-domain healthy training samples are used only in estimating $\boldsymbol{\sigma}_\delta$ and in the subsequent synthesis step, whereas real target-domain fault samples remain reserved for evaluation.

The discriminator uses binary cross-entropy with logits. The generator objective is

$$\mathcal{L}_G = \mathcal{L}_{\mathrm{BCE}} + 100\,(\mathcal{L}_{\mathrm{time}} + \mathcal{L}_{\mathrm{STFT}}), \tag{S61}$$

where $\mathcal{L}_{\mathrm{time}}$ is the mean absolute error between each generated sample and the perturbed simulated source-domain fault sample $\mathbf{Y}_S^c$ of the selected type, and $\mathcal{L}_{\mathrm{STFT}}$ is the mean absolute difference between their STFT magnitudes.

The STFT uses a 64-point Hann window, a 32-point hop, and normalized FFT. This spectral loss is computed over the resampled sample index and is not expressed in physical hertz. The generator and discriminator are optimized separately with Adam at a fixed learning rate of 10^{-4}, default coefficients $(\beta_1, \beta_2) = (0.9, 0.999)$, batch size 8, and at most 2,000 update steps. The generator is evaluated every 100 updates using fault-class-balanced source-validation MAE, and the checkpoint with the lowest MAE is retained. Early stopping is enabled after 600 updates, with a patience of six validation checks.

EHSS Step 4: Synthesize target fault-like samples

The target healthy training samples are aligned to the simulated healthy reference before generation. The generated residual is then transferred back to each target healthy sample:

$$\begin{aligned} \widetilde{\mathbf{X}}_T &= \mathbf{X}_T - \boldsymbol{\mu}_T + \boldsymbol{\mu}_S, \\ \widehat{\mathbf{Y}}_T^{\,c} &= \mathbf{X}_T + G\Big(\mathrm{cat}[\widetilde{\mathbf{X}}_T + \boldsymbol{\delta}, \mathbf{e}_c]\Big) - \Big(\widetilde{\mathbf{X}}_T + \boldsymbol{\delta}\Big) \end{aligned} \tag{S62}$$

Here $\boldsymbol{\mu}_S$ and $\boldsymbol{\mu}_T$ are the source and target healthy training means at the corresponding excitation frequency, and all signals are expressed in the fixed, channel-scaled coordinates. Conditions $c = 1, 2, 3, 4$ generate LVDT drift, LVDT degradation, solenoid degradation, and spool wear, respectively. Real target healthy samples provide class 0. No separate healthy-generation condition is required. The generated labels come from the fault-type code, not from actual target fault observations.

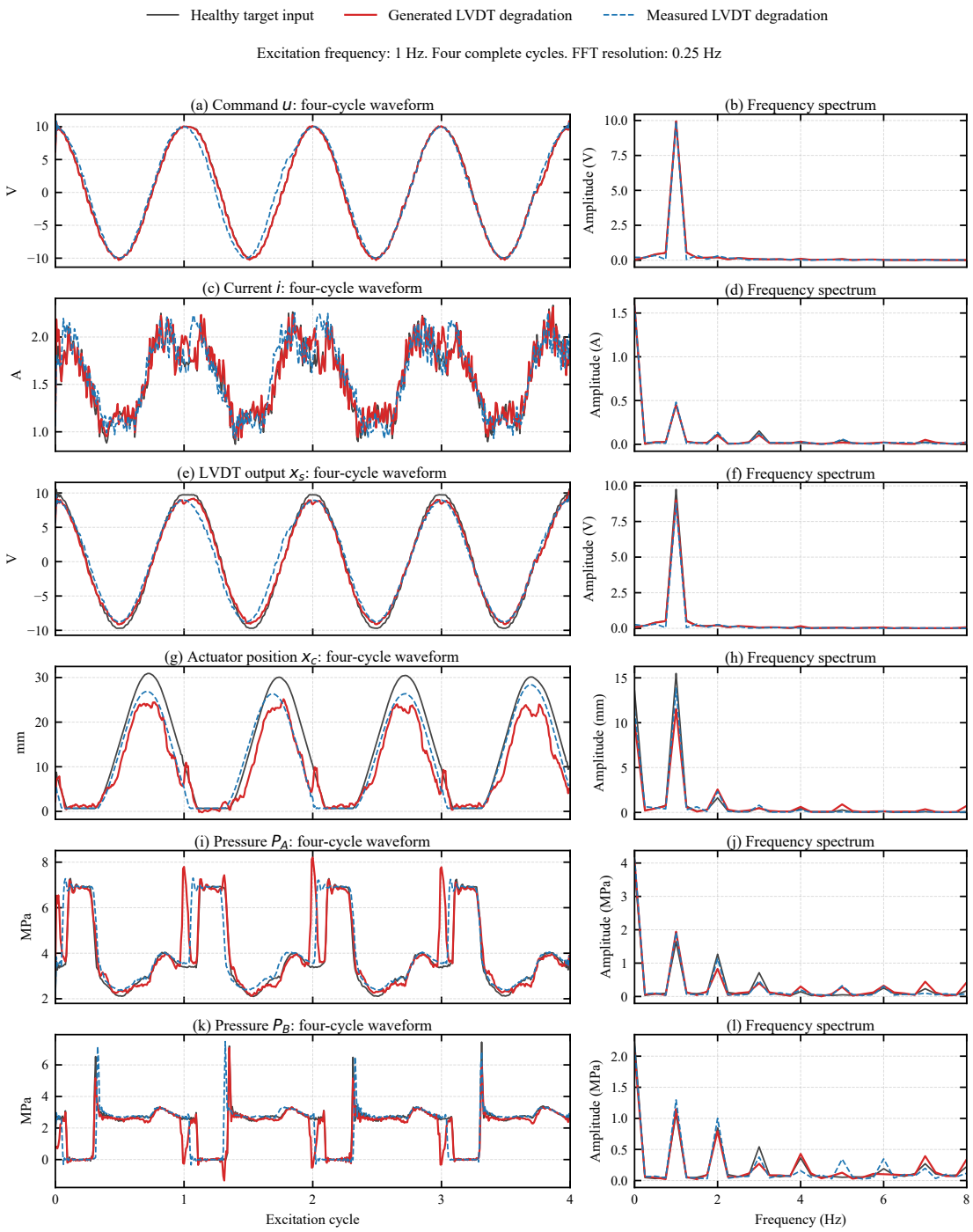


Fig. S16: Six-channel synthesis of an LVDT-degradation sample at 1 Hz.

EHSS Step 5: Train the target Self-ONN detector

The detector uses simulated source data together with real target healthy and generated target fault-like data. It has five third-order operational layers, each with 16 channels and kernel lengths $(9, 7, 5, 3, 3)$ at stride 2, followed by adaptive average pooling and dense layers of widths 32 and 5. Hidden activations use tanh. The five output logits are trained with cross-entropy. Adam uses a learning rate of 10^{-4}, default coefficients $(0.9, 0.999)$, and batch size 8. Training runs for at most 350 epochs, with 60 updates per epoch. The first 30 epochs use simulated source data only. Subsequent batches mix simulated source and target-domain training data in equal proportions, with balanced class

sampling.

The detector is evaluated every 10 epochs using only source-domain validation data. Its selection score is the macro-averaged recall across the five source classes. Ties are resolved by the lower source-validation cross-entropy. Only checkpoints after the source-only warm-up are eligible. Early stopping is enabled after 100 epochs, with a patience of eight validation checks. The selected checkpoint is used for evaluation, and the largest of its five logits determines the class in Eq. (S59).

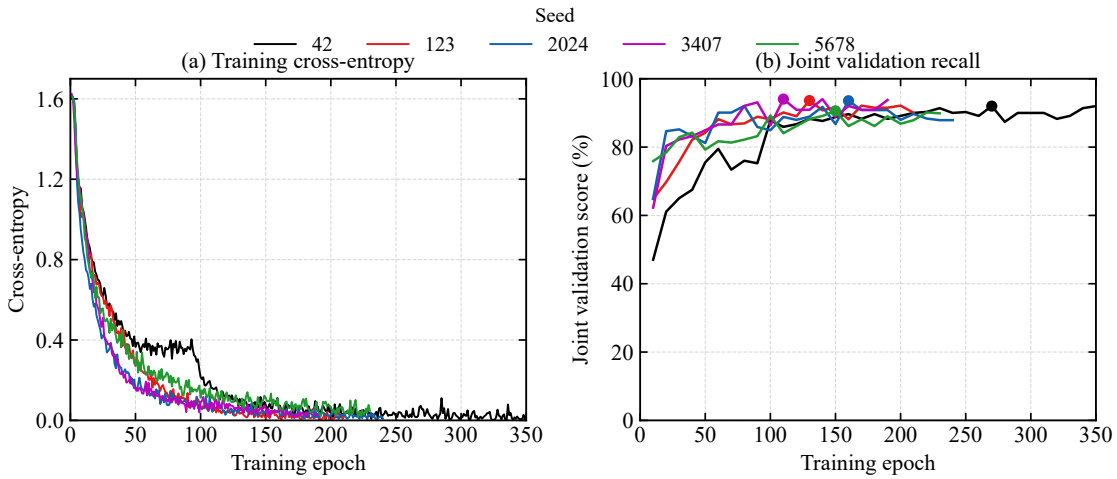


Fig. S17: Training cross-entropy and source-domain validation recall of the BDT target classifier over five seeds.

EHSS Step 6: Evaluate sample-wise classification

Testing is performed independently for each held-out healthy or fault sample. No record-level aggregation is applied. A 5×5 confusion matrix records the true and predicted classes, including cross-fault misclassification. The detector is trained with five seeds (42, 123, 2024, 3407, and 5678) on identical data partitions. Target faults are not used for checkpoint selection. Seed repetitions quantify training variability, not independent hardware acquisitions.

Table S7 summarizes the configuration specified for the EHSS-adapted BDT implementation, including its data representation, network structures, optimization settings, and evaluation rule.

Table S7: Configuration of the EHSS-adapted multiclass BDT implementation.

Item	EHSS-adapted implementation in this study
Domain and input	One EHSS system. The source consists of simulated healthy and faulty data spanning different excitation frequencies and fault types, while target-domain healthy samples are used for detector construction and target fault samples are held out for testing. Six-channel, 4-s, 4000-point samples are resampled to 512 points. Scaling is fixed from healthy training data.
Generator	1-D U-Net with 10-channel input (six signals plus four fault-type code channels). Encoder output widths $(32, 64, 128, 256, 512)$ and decoder output widths $(256, 128, 64, 32, 6)$. Five layers each, with skip connections and $Q = 3$.
Generator convolution	Kernel length 4, stride 2, padding 1 throughout, with tanh hidden activation and linear final output.
Discriminator	16-channel concatenation of healthy input, candidate fault, and class code. Five operational layers with widths $(16, 32, 64, 64, 64)$, kernel 4, stride 2, followed by a one-channel output layer with kernel 3, stride 1. Patch logits are averaged.

GAN objective	BCE-with-logits adversarial loss plus time MAE and STFT-magnitude MAE, weighted by 100. A 64-point Hann window, a 32-point hop, and a normalized FFT are used.
Input noise and perturbation	$\boldsymbol{\delta} = \boldsymbol{\sigma}_\delta \odot \mathbf{z}$, $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, with $\boldsymbol{\sigma}_\delta$ estimated from target healthy training residuals. The same $\boldsymbol{\delta}$ is added to the input and the supervising fault sample in training and to the input in synthesis.
GAN optimization	Separate Adam optimizers use a fixed learning rate of 10^{-4}, batch size 8, and at most 2,000 updates, with $(\beta_1, \beta_2) = (0.9, 0.999)$. Fault-class-balanced source-validation MAE is checked every 100 updates. The best-MAE checkpoint is selected, and early stopping is enabled after at least 600 updates with patience 6.
Target detector	Five operational layers, width 16, kernels $(9, 7, 5, 3, 3)$, stride 2, followed by adaptive average pooling and dense widths 32 and 5, with $Q = 3$, tanh hidden activations and five output logits.
Detector optimization	Adam uses a learning rate of 10^{-4}, five-class cross-entropy, batch size 8, and at most 350 epochs, with 60 updates per epoch. Its coefficients are $(\beta_1, \beta_2) = (0.9, 0.999)$. A 30-epoch source-only warm-up is followed by equal source/target mixing. Checkpoint selection uses source validation evaluated every 10 epochs. Early stopping is enabled after at least 100 epochs with patience 8.
Decision and evaluation	Each EHSS sample is classified independently as healthy, LVDT drift, LVDT degradation, solenoid, or spool fault by five-logit argmax. No record aggregation is used. Target fault samples are held out until testing.

S6.2 AFMTL-OCSVM

S6.2.1 Original method: experimental structures and workflow

Heravi et al. [2] evaluate two transfer tasks using measured acceleration: from the physical Qatar University Grandstand Simulator (QUGS) steel frame to either the Yellow Frame, a one-third-scale four-story steel frame, or the Z24 concrete bridge in Switzerland. An OC-SVM learns the healthy QUGS boundary, while an LSTM-based autoencoder (AE) maps healthy target spectra to healthy QUGS references. The fixed mapping and source detector then classify new target samples as healthy or damaged. Target damage is used only for evaluation. Thus, the method transfers a healthy reference rather than matching damage labels across structures. Figure S18 and Table S8 summarize the workflow and data.

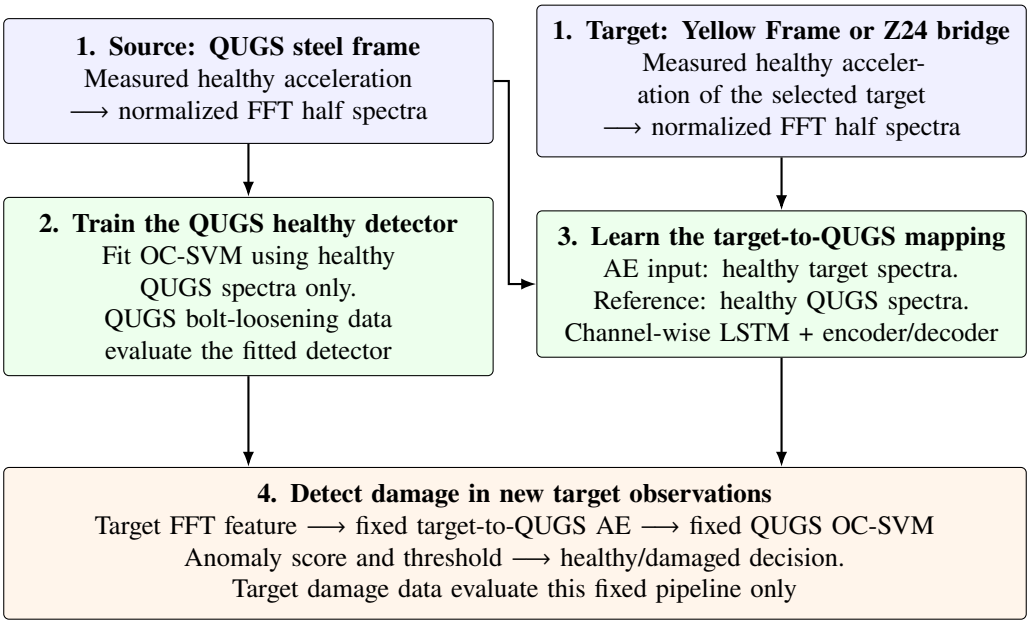


Fig. S18: Original structural damage-detection workflow from [2].

S6.2.2 Original method, Step 1: Construct spectral inputs for the three structures

For QUGS and the selected target structure, measured acceleration is divided into samples containing W time points. A normalized FFT is applied to each channel, and the half-spectrum amplitudes are concatenated across C channels. Denoting the per-channel feature extraction by $\mathcal{F}_W$, the input is written as

$$\mathbf{z} = \mathrm{col}[\,\mathcal{F}_W(\mathbf{x}_1), \ldots, \mathcal{F}_W(\mathbf{x}_C)\,], \qquad d = CW/2, \tag{S63}$$

where d follows the half-spectrum convention stated in the source paper. The principal setting is $W = 1000$. QUGS has 30 measured channels, whereas Yellow Frame and Z24 use 15. For transfer, the source channel count is reduced to 15 by retaining either the even- or the odd-indexed channels. This gives matching source and target feature dimensions of 7500. The unreduced QUGS feature dimension is 15000. Matching W and C matches the feature dimensions, not necessarily the physical record duration, because the structures have different sampling rates.

Table S8: Experimental datasets and principal feature settings in the original AFMTL-OCSVM study [2].

Structure and role	Sampling rate	Channels and feature size	Damage cases
QUGS steel frame, source	1000 Hz	30 channels, $d = 15000$, reduced to 15 channels, $d = 7500$, after matching	Healthy condition and 30 bolt-loosening conditions (DC1–DC30).
Yellow Frame steel structure, target	1000 Hz	15 channels, $d = 7500$	Healthy condition and 20 brace-removal damage conditions.
Z24 concrete bridge, target	100 Hz	15 reliable channels, $d = 7500$	Healthy baseline and 14 controlled damage conditions (DC1–DC14), including settlement, spalling, anchor-head failure, and tendon rupture.

S6.2.3 Original method, Step 2: Train and validate the QUGS healthy detector

The source OC-SVM is trained on the healthy QUGS training subset. The QUGS bolt-loosening samples evaluate whether this fixed boundary can detect departures from the healthy condition. They are not positive-class training examples for a supervised damage classifier. Given healthy source features $\{\mathbf{z}_{s,i}^0\}_{i=1}^{n_s}$, the one-class optimization is

$$\begin{aligned} \min_{\mathbf{w},\boldsymbol{\xi},\rho} \quad & \frac{1}{2}\|\mathbf{w}\|_2^2 + \frac{1}{\nu n_s}\sum_{i=1}^{n_s}\xi_i - \rho, \\ \text{subject to} \quad & \mathbf{w}^\top \Phi(\mathbf{z}_{s,i}^0) \geq \rho - \xi_i, \qquad \xi_i \geq 0, \end{aligned} \tag{S64}$$

where Φ is the kernel feature map and ν controls the one-class boundary. Its signed decision score is

$$s(\mathbf{z}) = \sum_{i=1}^{n_s} \lambda_i K(\mathbf{z}, \mathbf{z}_{s,i}^0) - \rho. \tag{S65}$$

Positive scores denote healthy inliers.

S6.2.4 Original method, Step 3: Map healthy target spectra to QUGS

For each target, channel-wise LSTMs of hidden width 100 feed a shared encoder and decoder. The complete mapping G_θ aligns healthy target features with the source reference:

$$G_\theta(\mathbf{z}_t^0) \approx \mathbf{z}_s^0. \tag{S66}$$

The source paper's Eq. (8) averages squared differences between each target sample and all source samples, allowing unequal sample counts. Writing the mapped output explicitly, this objective takes the form

$$\mathcal{L}_{\text{map}}(\theta) = \frac{1}{n_t n_s}\sum_{i=1}^{n_t}\sum_{j=1}^{n_s}\left\|G_\theta(\mathbf{z}_{t,i}^0) - \mathbf{z}_{s,j}^0\right\|_2^2. \tag{S67}$$

Here n_t and n_s are the target and source healthy sample counts. The original objective uses all target–source combinations rather than requiring matched one-to-one observations. The Z24 mapping experiment uses its healthy training subset as AE inputs.

S6.2.5 Original method, Step 4: Detect damage in Yellow Frame or Z24

At inference, a target feature is mapped by the fixed G_θ and evaluated by the source OC-SVM in Eq. (S65). The original paper also reports a healthy-score threshold

$$T = \mu + 3\sigma, \tag{S68}$$

where μ and σ are fitted from healthy training scores. An upper threshold requires a score that increases with abnormality. For the signed inlier score in Eq. (S65), the corresponding anomaly orientation is $a = -s$. This distinguishes the score sign from the threshold magnitude when implementing the decision rule.

Each damage condition is evaluated against healthy data as a binary problem, not as a damage-subtype class.

S6.2.6 EHSS adaptation: overview and workflow

The EHSS adaptation retains FFT features and healthy target-to-source mapping, but replaces the single healthy OC-SVM with four class-specific source models for joint classification. Figure S19 shows the processing chain, while Table S9 collects the implementation settings.

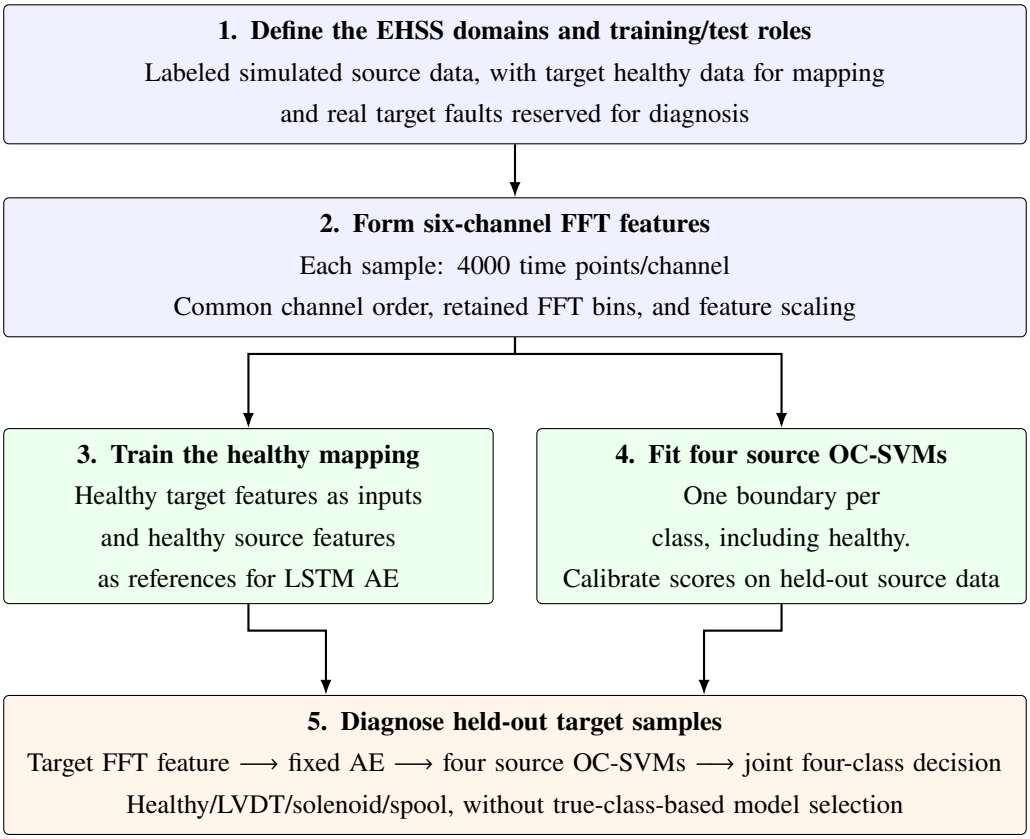


Fig. S19: Processing chain for adapting AFMTL-OCSVM to EHSS data.

S6.2.7 EHSS adaptation, Step 1: Define source and target data

For this AFMTL-OCSVM implementation, the four-class set remains

$$C = \{0 : \text{Healthy},\ 1 : \text{LVDT},\ 2 : \text{Solenoid},\ 3 : \text{Spool}\}. \tag{S69}$$

Let $\mathcal{D}_s^c$, $c \in C$, denote labeled simulated source samples, $\mathcal{D}_t^0$ the experimental target healthy samples, and $\mathcal{D}_t^f$ the target fault-test samples. Healthy target data are mapped to healthy source references, while labeled source data fit the class-specific OC-SVMs. LVDT drift and degradation share class 1. Target faults are reserved for final evaluation and excluded from training and score calibration.

Source training/calibration and target healthy training/test partitions are fixed across seeds. The partitions are formed from complete acquisition records, with no record shared between sets, and the 4-s signal segments are non-overlapping.

S6.2.8 EHSS adaptation, Step 2: Extract FFT features from each sample

Write a six-channel EHSS sample as

$$\mathbf{X} = [\mathbf{u}, \mathbf{i}, \mathbf{x}_s, \mathbf{x}_c, \mathbf{P}_A, \mathbf{P}_B] \in \mathbb{R}^{4000\times 6}. \tag{S70}$$

For channel c, a normalized amplitude spectrum can be represented by

$$A_c[k] = \frac{1}{4000}\left|\sum_{n=0}^{3999} X[n,c]\exp\left(-\frac{2\pi \mathrm{i} k n}{4000}\right)\right|, \tag{S71}$$

with i denoting the imaginary unit. Given retained-bin set $\mathcal{B}$ and feature-scaling operator $\mathcal{S}$, the model feature is

$$\mathbf{z} = \mathcal{S}\Big(\mathrm{col}_{c=1}^{6}\{A_c[k] : k \in \mathcal{B}\}\Big), \qquad d_{\mathrm{EHSS}} = 6|\mathcal{B}|. \tag{S72}$$

The revised implementation retains bins $\mathcal{B} = \{0, \ldots, 1999\}$, including DC and excluding Nyquist, with no FFT padding. This gives 12000 features. Channel-wise RMS scaling is fitted on healthy source training spectra and fixed for all source and target samples.

Retaining DC and using fixed training-derived scaling preserve the offset and amplitude changes associated with LVDT faults.

S6.2.9 EHSS adaptation, Step 3: Train the LSTM mapping with healthy data

The residual LSTM–AE mapping is

$$\begin{aligned} \mathbf{h}_c &= \mathrm{LSTM}_{\theta_c}(\mathbf{z}_{t,c}^0), \qquad c = 1, \ldots, 6, \\ G_\theta(\mathbf{z}_t^0) &= \mathbf{z}_t^0 + D_{\theta_D}\left(E_{\theta_E}(\mathrm{col}[\mathbf{h}_1, \ldots, \mathbf{h}_6])\right), \end{aligned} \tag{S73}$$

where E_{θ_E} and D_{θ_D} denote the encoder and decoder. The LSTMs operate on spectral features, not raw time sequences. Each channel's 2000-bin spectrum is split into 100 consecutive tokens of width 20, which form the LSTM input sequence. An identity skip and the following regularizer limit collapse toward a constant healthy reference:

$$\mathcal{L}_{\mathrm{map}}^{\mathrm{EHSS}} = \frac{1}{d_{\mathrm{EHSS}}}\left[\mathcal{L}_{\mathrm{map}} + \frac{\lambda_{\mathrm{id}}}{n_t}\sum_{i=1}^{n_t}\|G_\theta(\mathbf{z}_{t,i}^0) - \mathbf{z}_{t,i}^0\|_2^2\right], \qquad \lambda_{\mathrm{id}} = 1. \tag{S74}$$

Both terms use feature-wise mean squared errors. Expanding the square gives $\frac{1}{n_s}\sum_j \|g - \mathbf{z}_{s,j}^0\|_2^2 = \|g - \bar{\mathbf{z}}_s^0\|_2^2 + \text{const}$, so the all-pairs term is equivalent to regressing each mapped target sample onto the source healthy mean $\bar{\mathbf{z}}_s^0$, which

is the source-moment reduction used in the implementation. Without the identity term this collapses the mapping toward a constant. With $\lambda_{\mathrm{id}} = 1$, the per-sample unconstrained minimizer is $(\mathbf{z}_t^0 + \bar{\mathbf{z}}_s^0)/2$. Network dimensions, initialization, and optimization settings appear in Table S9.

Figure S20 shows the training-loss history for seeds 42, 123, 2024, 3407, and 5678. Each run completes 2525 optimizer updates. The initial decrease is followed by fluctuations at a lower loss level, reflecting the mini-batch optimization process.

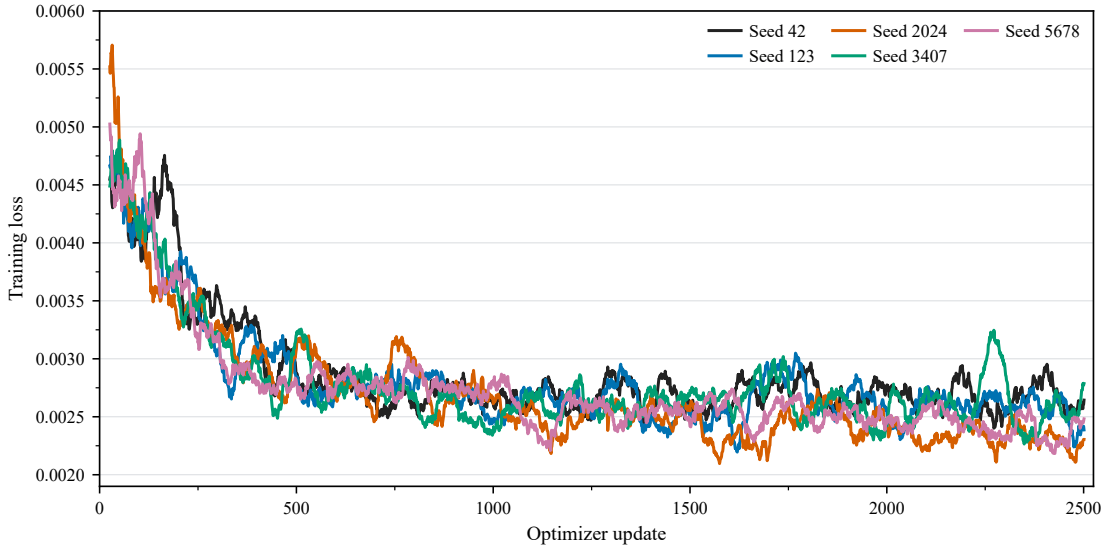


Fig. S20: Training loss in the additional five-seed AFMTL study.

Figure S21 evaluates the objective on a fixed training subset and on held-out healthy/source-calibration data. The checkpoint with the lowest held-out calibration objective is retained for each seed, even when it precedes the final update. The selected calibration objectives are 7.3%–8.8% below their initial values.

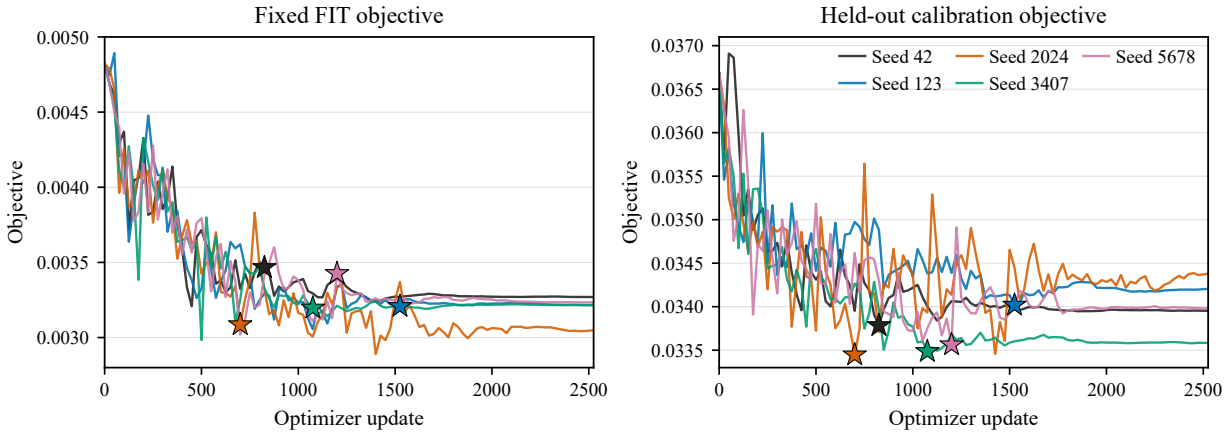


Fig. S21: Fixed training-subset (left) and held-out calibration (right) objectives in the additional AFMTL study.

S6.2.10 EHSS adaptation, Step 4: Fit and fix the class-specific OC-SVMs

One OC-SVM is fitted to each simulated source class using the optimization in Eq. (S64), with that class's samples replacing the healthy-only fitting set. All four models use an RBF kernel, $\nu = 0.05$, and $\gamma_c = 1/(d\ \mathrm{Var}(\mathbf{Z}_s^c))$,

where the variance is taken over all entries of the class-c training matrix (the `gamma=scale` setting), so γ_c differs between classes. For a mapped target sample, its class-c inlier score is

$$s_c(\mathbf{z}_t) = \sum_{i=1}^{n_s^c} \lambda_{c,i} K_c\left(G_\theta(\mathbf{z}_t), \mathbf{z}_{s,i}^c\right) - \rho_c. \tag{S75}$$

The scores are normalized using held-out source samples of the corresponding class. Let τ_c be the fifth percentile of their inlier scores and σ_c their sample standard deviation. The joint decision is

$$r_c(\mathbf{z}_t) = \frac{s_c(\mathbf{z}_t) - \tau_c}{\max(\sigma_c, 10^{-8})}, \qquad \widehat{y}_t = \arg\max_{c \in C} r_c(\mathbf{z}_t). \tag{S76}$$

Every mapped target sample is scored by all four models without access to its true label. The closed-set maximum-score rule also applies to overlapping or all-negative responses. Ties select the lowest class index. All-negative responses are additionally logged. These scores are not probabilities. This rule replaces the original binary threshold.

S6.2.11 EHSS adaptation, Step 5: Evaluate the fixed mapping and detector

Each held-out 4000-point target sample receives one label using Eq. (S76). Its class-wise recall is

$$\text{Recall}_c(\%) = \frac{N_{y=c,\widehat{y}=c}}{N_{y=c}} \times 100. \tag{S77}$$

A 4×4 confusion matrix includes healthy data and the three fault classes, with wrong-fault labels counted as errors. Overall accuracy and healthy false-alarm rate are also reported for each seed and operating condition. Binary detection rates cannot replace these multiclass results.

S6.2.12 EHSS implementation settings

Table S9 summarizes the network and implementation settings of the EHSS-adapted AFMTL-OCSVM.

Table S9: Network and implementation settings of the EHSS-adapted AFMTL-OCSVM.

Network/method setting	EHSS adaptation
Source and target domains	Simulated EHSS source and experimental target, with four classes defined in Eq. (S69).
Training data roles	Healthy source/target data train the mapping, while labeled simulated source data fit four class-specific OC-SVMs.
Target fault access	Target faults are reserved for final evaluation. No target fault labels enter training, score calibration, or model selection.
Source healthy split	Separate source training/calibration partitions, with all source classes following the same fixed partition protocol.
Target healthy split	Separate target healthy training/test partitions.
Sampling rate	1000 Hz for all six EHSS channels.
Time points per sample	4000 time points per channel (4 s at 1000 Hz), no FFT padding.
Channels	Six channels: u, i, x_s, x_c, P_A, P_B.
Channel matching	Common channel order u, i, x_s, x_c, P_A, P_B, with no channel removal.
Condition coverage	LVDT drift, LVDT degradation, proportional-solenoid degradation, and spool wear at 1, 5, 10, and 20 Hz as listed in the manuscript.
Feature type	Normalized FFT amplitudes using bins 0–1999, with DC retained and Nyquist omitted.
Feature dimension	12000 features, reshaped as six channels of 2000 bins.

Table S10: Network and implementation settings of AFMTL-OCSVM (continued: mapping network and training).

Network/method setting	EHSS adaptation
Feature scaling	Source-healthy-fit channel RMS scaling, frozen for all source and target samples.
Mapping direction	Healthy target $\longrightarrow$ healthy source, applied to all target test samples.
Channel encoder	One LSTM per channel with 100 spectral tokens of width 20.
LSTM hidden dimension	100 per channel, with six final hidden states concatenated to width 600.
Encoder/decoder schematic	Encoder $600 \to 100 \to 50$ and decoder $50 \to 100 \to 12000$, with an identity skip added.
Latent representation	Latent width 50, with the final residual-output layer initialized to zero.
Activations and regularization	tanh hidden layers and linear residual output, with no dropout or weight decay and an identity-loss weight of 1.
Mapping objective	Healthy all-pairs MSE plus identity-preservation MSE in Eq. (S74).
Mapping sample correspondence	All-to-all healthy association, implemented using the equivalent source-moment reduction.
AE optimizer	Adam, $(\beta_1, \beta_2) = (0.9, 0.999)$.
AE learning rate	Fixed 10^{-4}.
AE batch size	8.
AE training duration	200 fixed epochs, with the final checkpoint retained and gradient-norm clipping at 1.

Table S11: Network and implementation settings of AFMTL-OCSVM (continued: OC-SVM and evaluation).

Network/method setting	EHSS adaptation
OC-SVM fitting objective	Same one-class objective separately for each source class, giving four OC-SVMs in total.
OC-SVM kernel	RBF, `gamma=scale`, fitted within each source class.
OC-SVM ν	$\nu = 0.05$ for all four models.
OC-SVM score orientation	Class inlier scores normalized using held-out source class scores. The maximum normalized score selects the class.
Threshold rule	Per-class fifth-percentile inlier boundary and sample standard deviation from source calibration data, as defined in Eq. (S76).
Number of OC-SVMs	Four: healthy, merged LVDT, solenoid, and spool, with joint argmax without true-class-based model selection.
Inference sequence	FFT $\to$ fixed residual mapping $\to$ all four OC-SVMs $\to$ normalized-score argmax for each 4000-point sample.
Evaluation measures	Four-class confusion counts, overall classification accuracy, per-class recall, and healthy false-alarm rate, for every condition and seed.

S6.3 CNN-based ablation of the hybrid PI-RNN predictor

The CNN-based method is an ablation of the proposed framework. As specified for Method 3 in the manuscript, the hybrid PI-RNN is replaced by a data-driven CNN for normal-response prediction. The hierarchical inference, residual fusion, and fault-localization procedure are retained. This comparison evaluates the contribution of the hybrid PI-RNN predictor within the same diagnostic framework.

1. CNN Architecture and Training Procedure

To evaluate the contribution of physics-informed dynamic modeling, a CNN-based ablation model is constructed by replacing the hybrid PI-RNN with a data-driven CNN predictor, while retaining the remaining hierarchical multi-source diagnostic framework. The CNN is designed to learn the normal dynamic behavior of the EHSS and predict its subsequent physical responses, rather than directly classifying fault categories.

For each target operating condition, the CNN is trained using only the corresponding healthy experimental data. The same recording-level training and validation partition as the proposed method is adopted, ensuring that recordings are not shared between the two subsets. Fault samples are excluded from the predictor training process and are used only for subsequent diagnostic evaluation.

The CNN takes the available command and system-state information as input and extracts their relationships through convolutional feature extraction. The resulting features are mapped to a regression output containing five physical variables. Specifically, the one-step prediction is expressed as

$$\widehat{\mathbf{Y}}^{\mathrm{CNN}}_{k+1|k} = \mathcal{F}_{\boldsymbol{\theta}_{\mathrm{CNN}}}(\mathbf{Z}_k), \qquad \mathbf{Y}_{k+1} = [i_{k+1}, x_{s,k+1}, P_{A,k+1}, P_{B,k+1}, x_{c,k+1}]^{\mathrm{T}} \tag{S78}$$

where $\mathbf{Z}_k$ represents the command and state information available up to time step k, and $\boldsymbol{\theta}_{\mathrm{CNN}}$ denotes the trainable CNN parameters. The predicted variables and their channel ordering are identical to those of the hybrid PI-RNN, ensuring compatibility with the subsequent diagnostic framework.

During training, the CNN learns the normal input–output relationship by minimizing the prediction discrepancy between its outputs and the corresponding next-step healthy measurements. The training objective is expressed as

$$\boldsymbol{\theta}^{*}_{\mathrm{CNN}} = \arg\min_{\boldsymbol{\theta}_{\mathrm{CNN}}} \frac{1}{N} \sum_{k=1}^{N} \ell\left(\widehat{\mathbf{Y}}^{\mathrm{CNN}}_{k+1|k}, \mathbf{Y}_{k+1}\right), \tag{S79}$$

where N is the number of healthy training samples and $\ell(\cdot,\cdot)$ denotes the prediction loss. Unlike the hybrid PI-RNN, the CNN does not explicitly incorporate governing physical equations into its prediction process.

Five independent training runs are performed using random seeds 42, 123, 2024, 3407, and 5678. The data partitions remain unchanged across these runs. After training, the CNN parameters are fixed and the trained predictor is used to estimate the normal system response under the corresponding operating condition. The prediction is generated using only information available before the measurement being evaluated.

Retained hierarchical diagnosis and evaluation

To ensure a consistent comparison, only the PI-RNN prediction module is replaced by the CNN, while the hierarchical multi-source diagnostic framework remains unchanged. The CNN predictions are substituted into the model-based deviation calculation.

The upstream and downstream physical inferences remain unchanged. The two chamber pressures are retained as parallel signals at the same hydraulic level. The resulting deviations undergo the same normalization, saturation, threshold activation, and convolution operations using the fixed diagnostic kernels $\boldsymbol{K}_S$ and $\boldsymbol{K}_C$ as in the proposed method. These physics-designed diagnostic kernels are distinct from the trainable convolutional filters within the CNN predictor.

Sensor anomalies and component faults are identified and localized using the same hierarchical indicator patterns and decision rules. The confidence-weighted measurement fusion and subsequent prediction update are also retained, with the CNN providing the predicted physical responses.

Both methods are evaluated using the same fault samples, operating conditions, data partitions, and five random seeds. Consequently, the ablation comparison evaluates the effect of replacing physics-informed dynamic prediction with data-driven CNN prediction while preserving the hierarchical diagnostic structure.

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